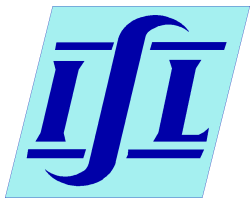
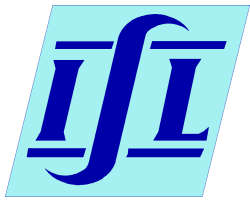


Introduction & Welcome



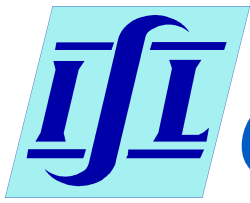
Data Mining - About this Course

- n This course will:
 - | Introduce the concepts and techniques of data mining.
 - | Give you hands on experience of data mining techniques.
 - | Teach you the basics of the Clementine/SPSS Modeller Data Mining System.
 - | Prepare you to identify, assess and execute data mining projects.



This Course will not

- n Cover data mining techniques in exhaustive detail.



Clementine Clementine/SPSS

Modeller Data Mining

Day 1:

Course Outline

- n Introduction
 - I ISL & Clementine - Background
 - I Data Mining Overview

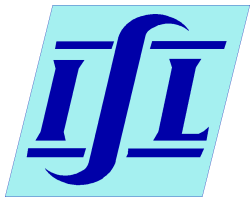
- n Data Exploration and Manipulation
 - I Workshop 1: Data sources
 - I Workshop 2: Exploration and manipulation
- n Modelling
 - I Neural Networks
 - I Rule Induction
 - I Workshop 4: Neural networks and rule induction



Clementine/SPSS Modeller Data Mining Course Outline

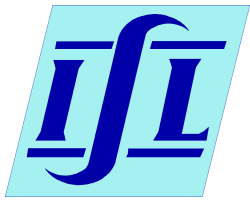
Day 2:

- n “Unsupervised” Modelling
 - l Clustering (using Kohonen Net)
 - l Workshop 5: Clustering
 - l Association (using Apriori and GRI)
 - l Workshop 6: Association detection
- n Data Mining Projects
 - l Workshop 7 - Discuss Data Mining Applications
- n Tips & Hints
- n Advanced features
- n Discussion with Consultants & Advanced Practicals



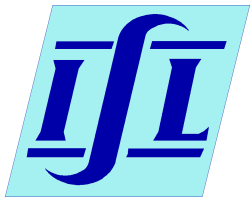
Integral Solutions Limited & Clementine/SPSS Modeller

Background



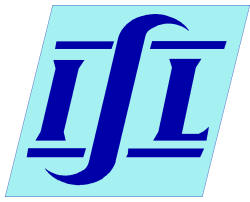
ISL

- n Formed 1989 in management buy-out from SD-Scicon
- n Supply of advanced software and services
- n Active in “data mining” from 1990
- n Presence in UK, USA (ISL DSI), Singapore (ISAPL)
 - l distributors around the world
- n Over 550 customers

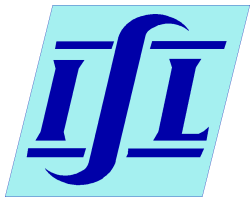


Clementine/SPSS Modeller

- n Development initiated in 1992
- n First commercial versions shipped Oct. 1994
 - l tool for business users, not just technologists
- n Users in finance, government, retail, manufacturing, pharmaceuticals, ...
- n Rated as market leader by analysts
- n Clementine V4 winner of Information Age Product of the Year Award, 1997



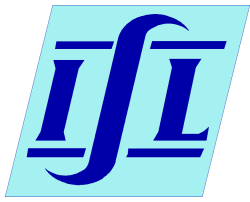
Data Mining Overview



What is Data Mining?

- n Data mining is:
 - | finding patterns in your data...
 - | ...which you can use...
 - | ...to do your business better

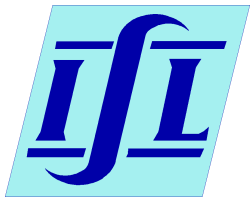
- n “Decisions from Data”



What is Data Mining?

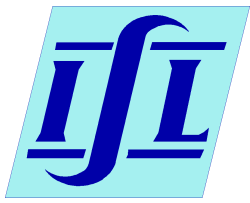
- n Data mining is not:
 - | throwing masses of data at smart technologies and getting magical answers

- n Data mining is:
 - | an interactive, iterative process, combining advanced analysis techniques and business expertise



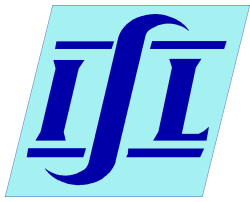
Data Mining with Clementine/SPSS Modeller

- n Clementine combines multiple tools and technologies to support the process of interactive data mining
- n Clementine makes data mining accessible to non-technologists, allowing business knowledge to be applied directly

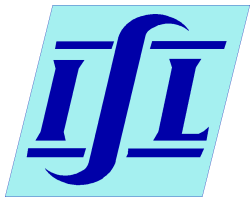


Data Mining with Clementine/SPSS Modeller

- n Clementine provides *passive* and *active* support for data mining
- n Passive support:
 - l Initiative comes from user
 - u e.g. visualisation, statistics
- n Active support:
 - l Computer plays a much more active role
 - u e.g. machine learning

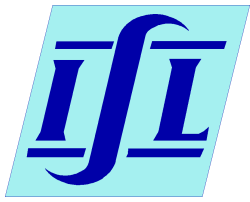


Data Exploration



Techniques

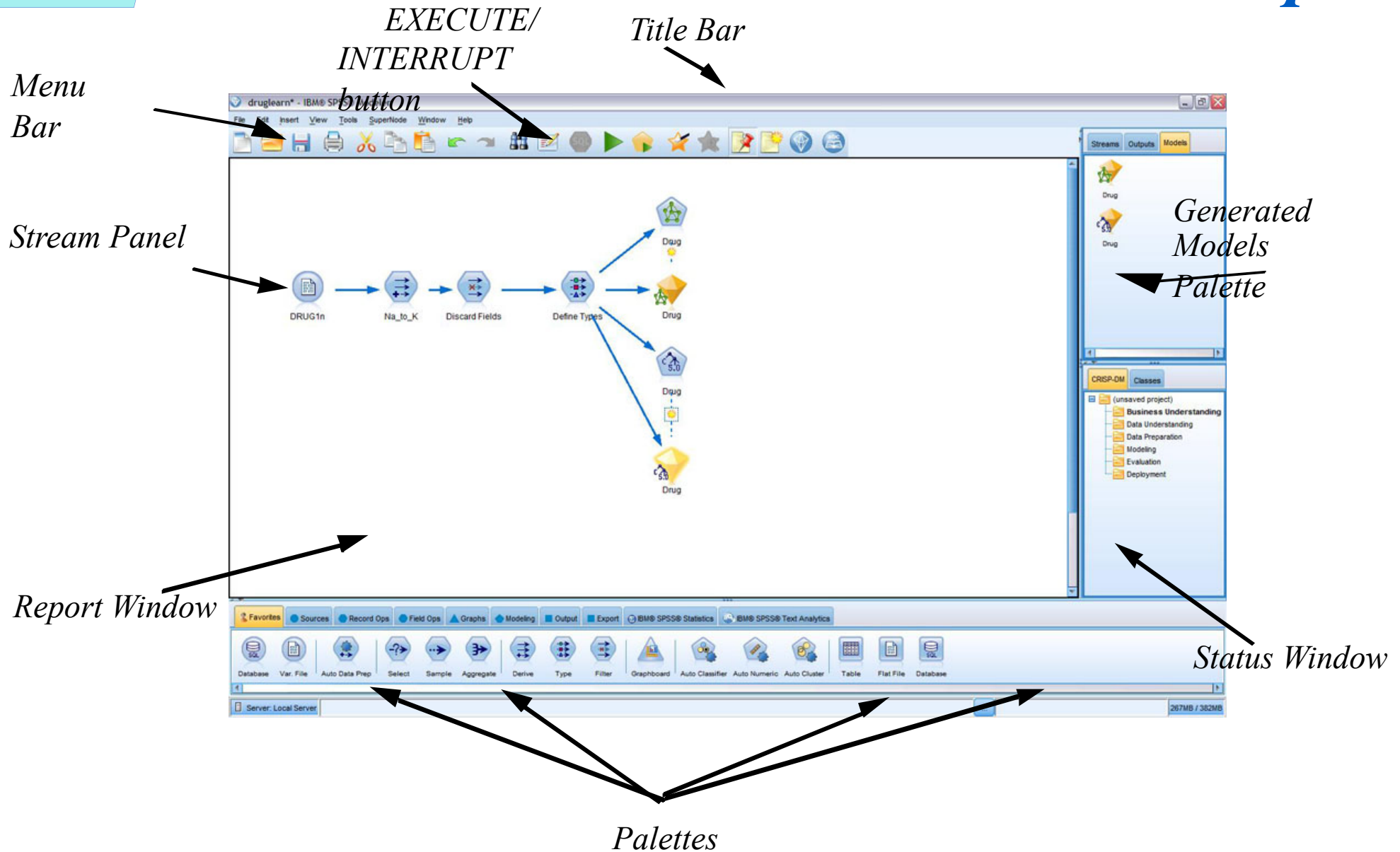
- n **“Passive” Computer Support**
 - | **Browsing & Visualisation**
 - | **Statistics**
 - | **Hypothesis testing**
- n **“Active” Computer Support**
 - | **Neural Networks**
 - | **Rule Induction**
 - | **Techniques combined**
 - | **Regression Modelling**



The Clementine/SPSS Modeller User Interface

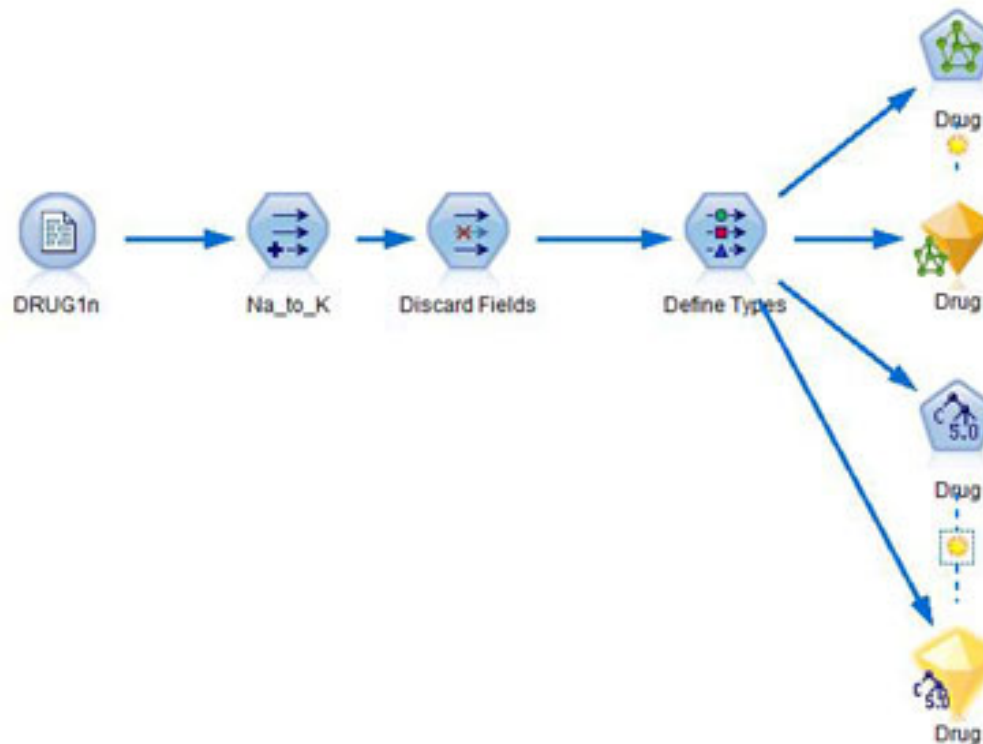


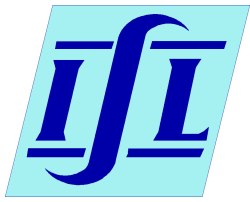
The Clementine/SPSS Modeller Desktop



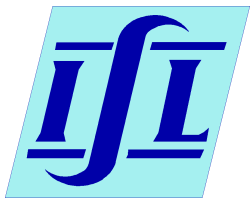
Visual Programming in Clementine/SPSS Modeller

- n Based on icons (= "nodes")
- n Pick icons from palette, place them, edit their attributes
- n Connect to specify flow of data ("streams")



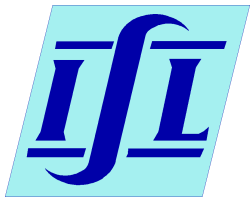


Data Sources & Data Sets



DBMS via ODBC

- n Industry standard open connectivity to many DBMS.
 - l ODBC - Open Data-Base Connectivity
- n Tables selected by name, or a database query used
 - specify SQL expression
 - e.g.
**SELECT CNAME, CHIST, PROFIT
FROM CUST, TRANS
WHERE CUST.CID = TRANS.CID**
- n Access via DBMS
- n Prompted for authorisation
- n Names of fields, types etc. from data dictionary



Var. File



Fixed File

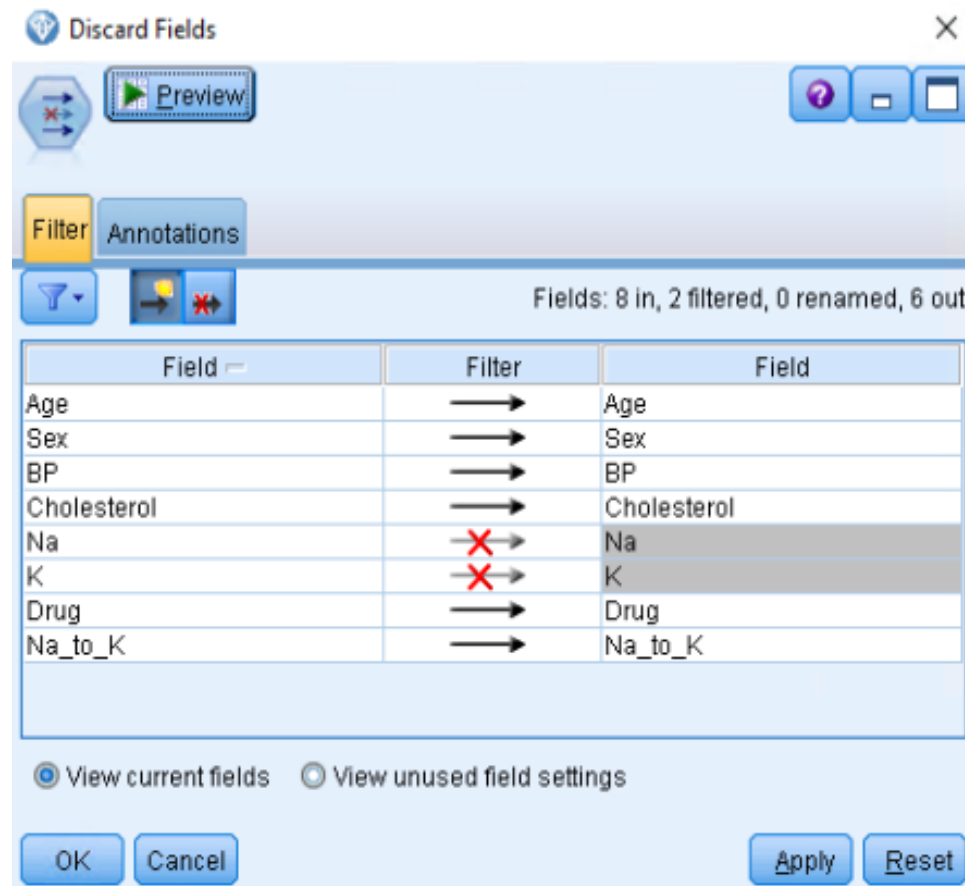
ASCII Sources

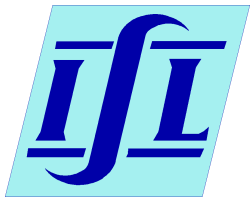
- n ASCII files (Portability & Legacy Data)
- n "Fixed" file format
 - all records have same length
 - field defined by position/length
- n "Variable" file format
 - delimiter based
 - specify delimiter structure for parsing
- n Where possible, fields are converted to numbers (may be turned off)

Filter Nodes



- n Specify which fields are to be used/ignored
- n Allow (re)naming of fields





Browsing & Visualisation

- n Viewing properties of variables and relationships between variables
- n Range from tabulation to multi-dimensional plots
- n In Clementine:
 - Tables
 - Distributions
 - Plots (+ overlays)
 - Histograms
 - Collections
 - Webs
 - Matrices

Tables



Table

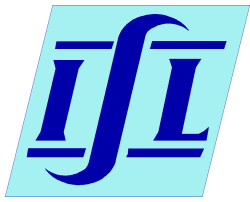
- n Present data as text, record by record
- n "Spreadsheet" style display
- n Can interact with table (e.g. highlighting, selecting subset of records)

Table (7 fields, 200 records)

File Edit Generate

Table Annotations

	Age	Sex	BP	Cholesterol	Na	K	Drug
1	23	F	HIGH	HIGH	0.793	0.031	drugY
2	47	M	LOW	HIGH	0.739	0.056	drugC
3	47	M	LOW	HIGH	0.697	0.069	drugC
4	28	F	NORMAL	HIGH	0.564	0.072	drugX
5	61	F	LOW	HIGH	0.559	0.031	drugY
6	22	F	NORMAL	HIGH	0.677	0.079	drugX
7	49	F	NORMAL	HIGH	0.790	0.049	drugY
8	41	M	LOW	HIGH	0.767	0.069	drugC
9	60	M	NORMAL	HIGH	0.777	0.051	drugY
10	43	M	LOW	NORMAL	0.526	0.027	drugY
11	47	F	LOW	HIGH	0.896	0.076	drugC
12	34	F	HIGH	NORMAL	0.668	0.035	drugY
13	43	M	LOW	HIGH	0.627	0.041	drugY
14	74	F	LOW	HIGH	0.793	0.038	drugY
15	50	F	NORMAL	HIGH	0.828	0.065	drugX
16	16	F	HIGH	NORMAL	0.834	0.054	drugY

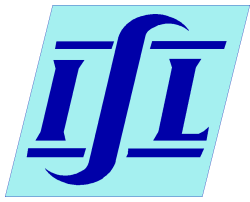


Workshop 1

n Data sources

n Aim

- | Connect to data source
- | View data via tables
- | Filter fields in data



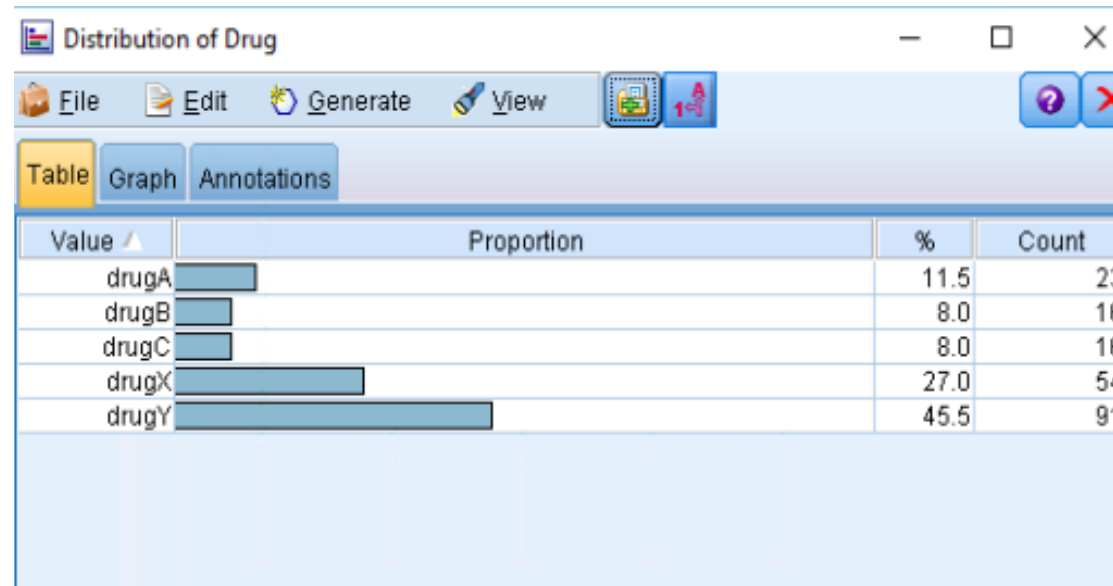
Summary

- n Can connect to data in databases/files
- n Filter node controls field inclusion/exclusion
- n Tables provide simple display of “raw” data

Distribution

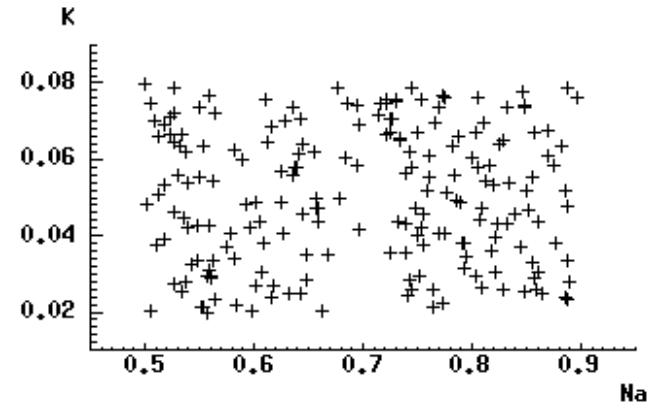


- n Shows distribution of symbolic values (e.g. drug)
- n Can overlay another symbolic field (e.g. BP)

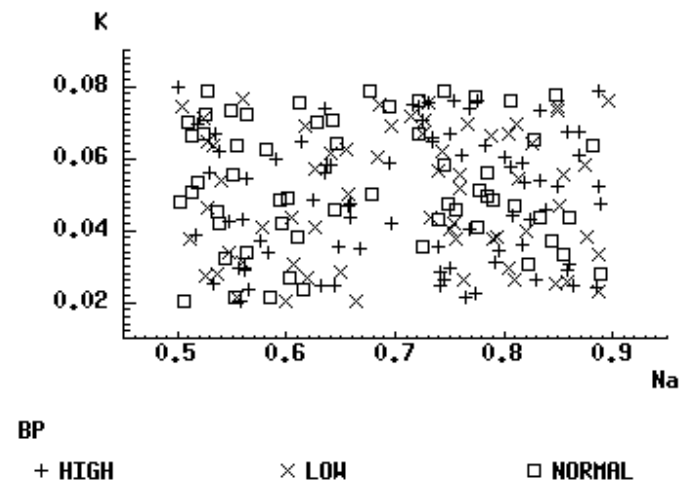


- n Typical uses:
 - | Compare "profiles" of different (sub) populations
 - | Identify imbalances in data sets

Plots

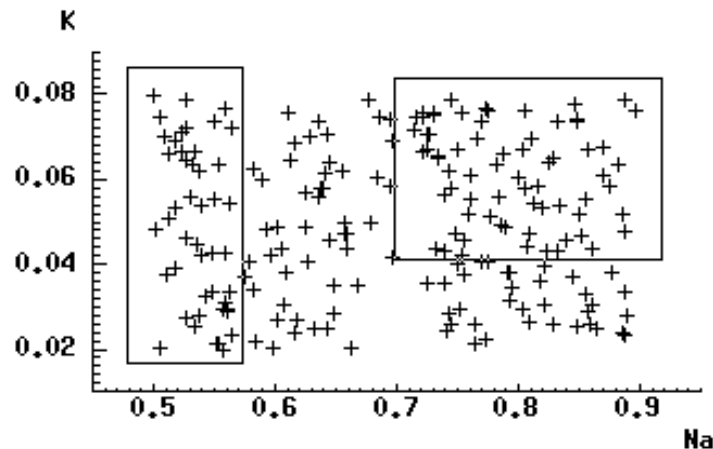


- n Point or line (various styles)
- n Used to show relationship between 2 numeric attributes
- n Can overlay third (symbolic) attribute

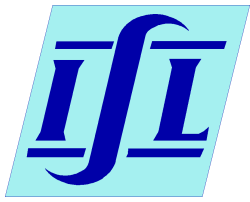


Interacting with Plots

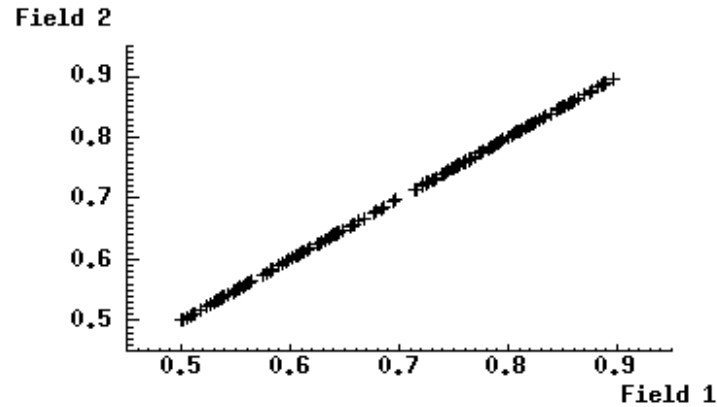
- n Can define regions of interest with mouse:



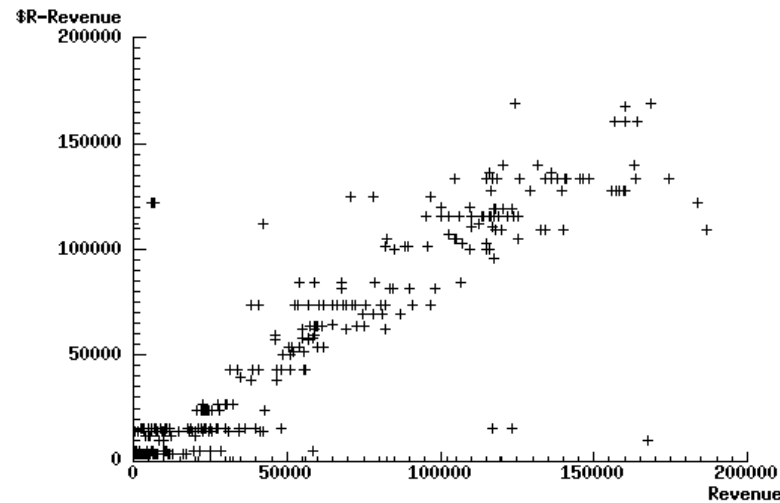
- n region defined by ranges e.g. $0.5 < x < 1.0$
 $11.7 < y < 25.8$
- n Facilities:
 - | select records on the basis of being in one or all regions
 - | create a flag to indicate a record is in one or all regions
 - | add a field which indicates which region (if any) a record is in
- n Often use region selection to pick up outliers and clusters



Some interesting graphs (1)

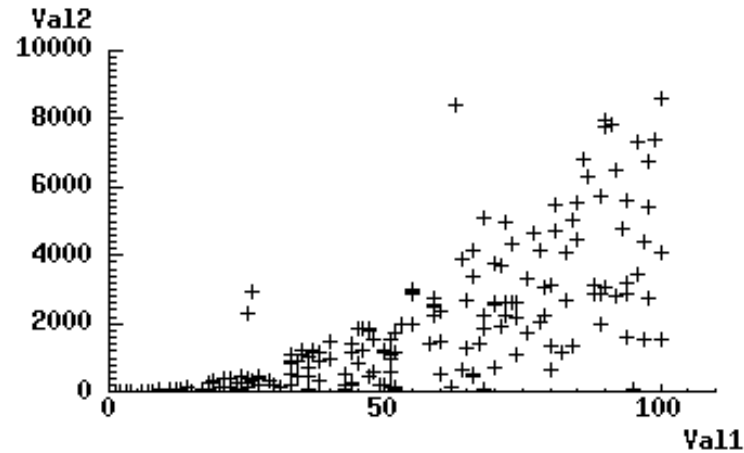


(Unbelievably) perfect relationship

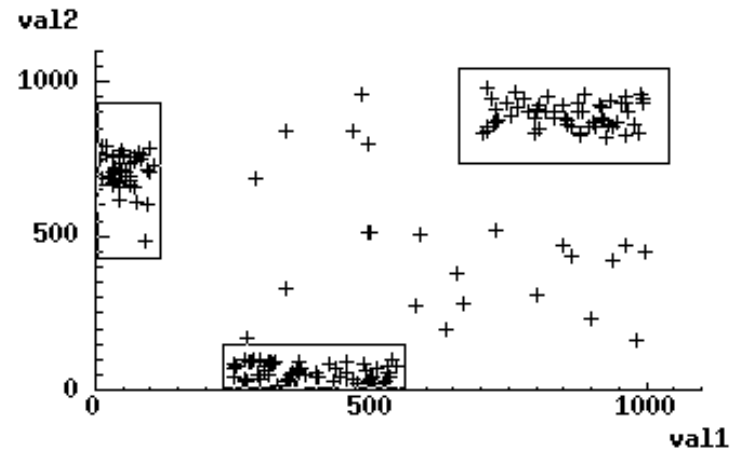


Reasonably good prediction/correlation

Some interesting graphs (2)

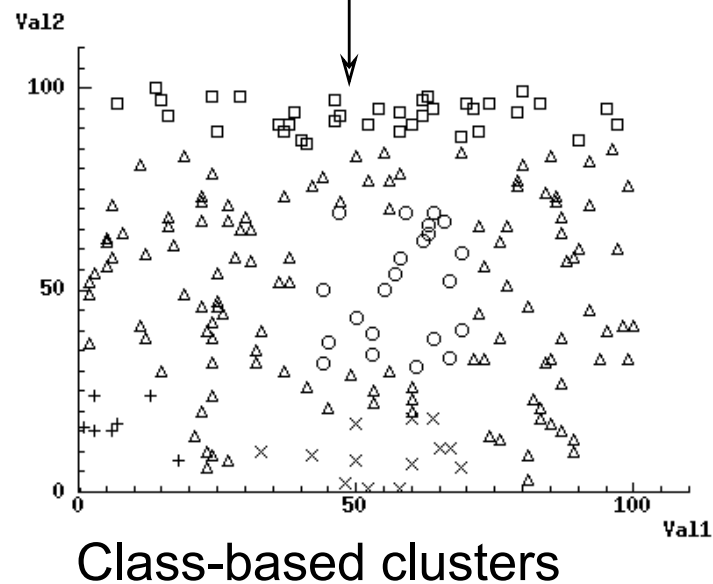
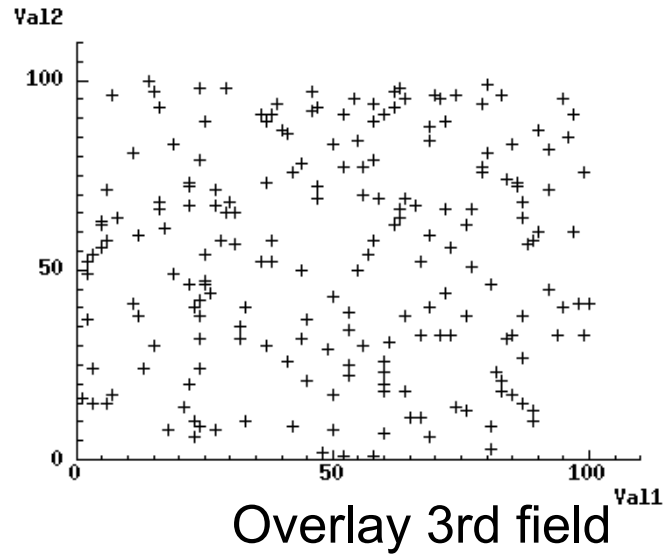


Apparent relationship, with some outliers

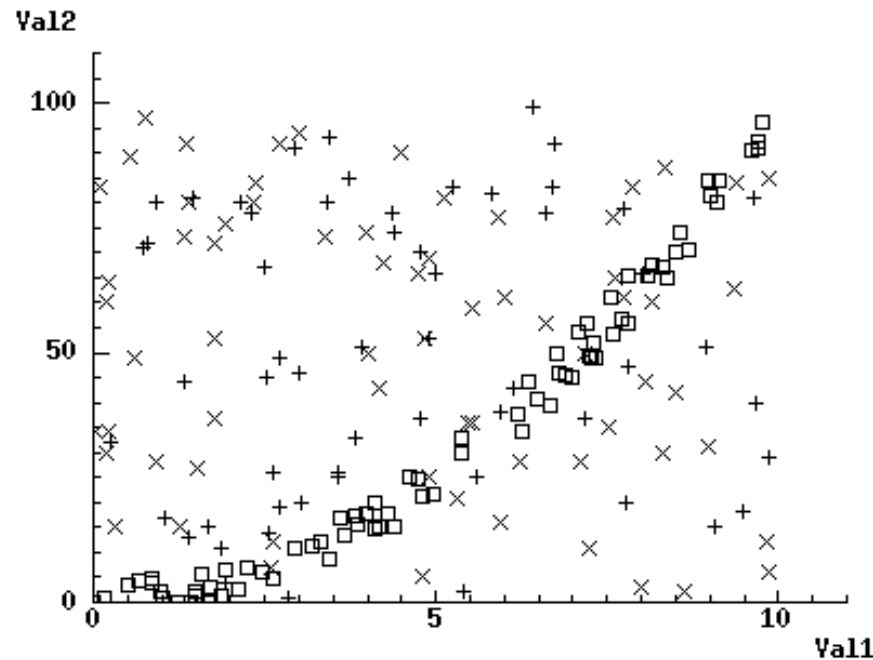


Clusters (shown in selected regions)

Some interesting graphs (3)

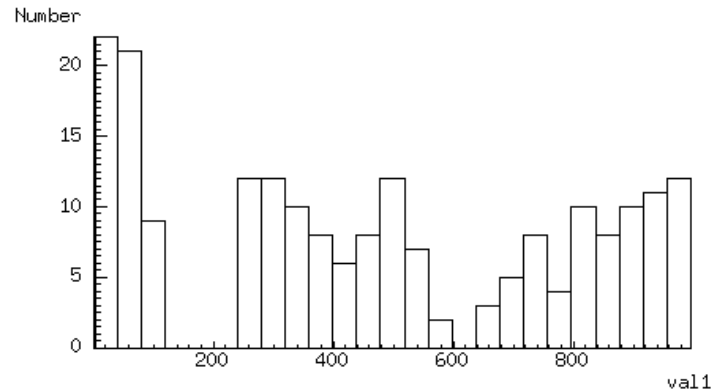


Some interesting graphs (4)

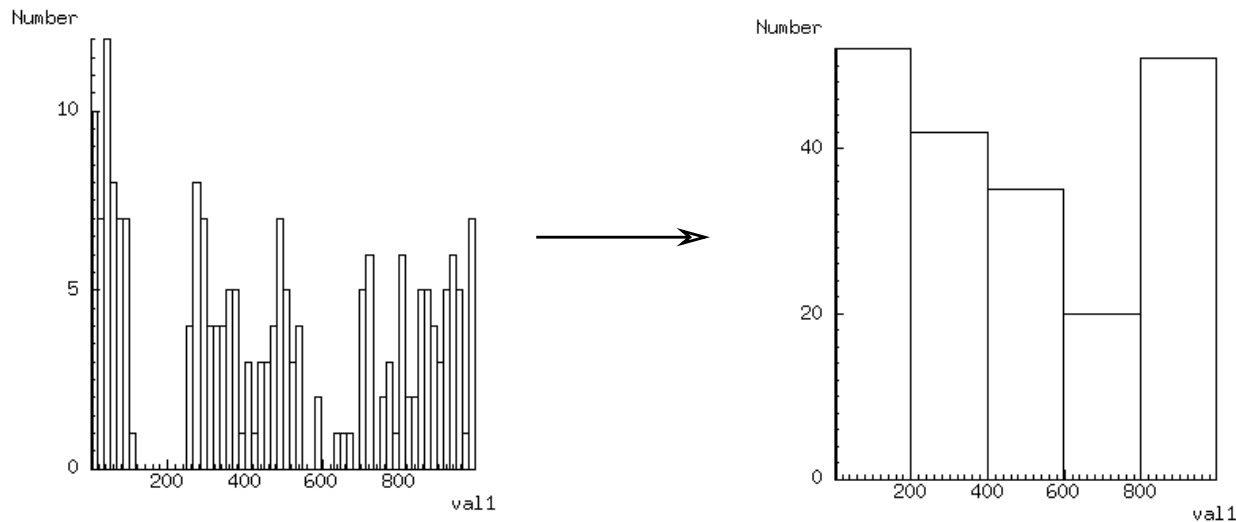


Relationship in one class, not others

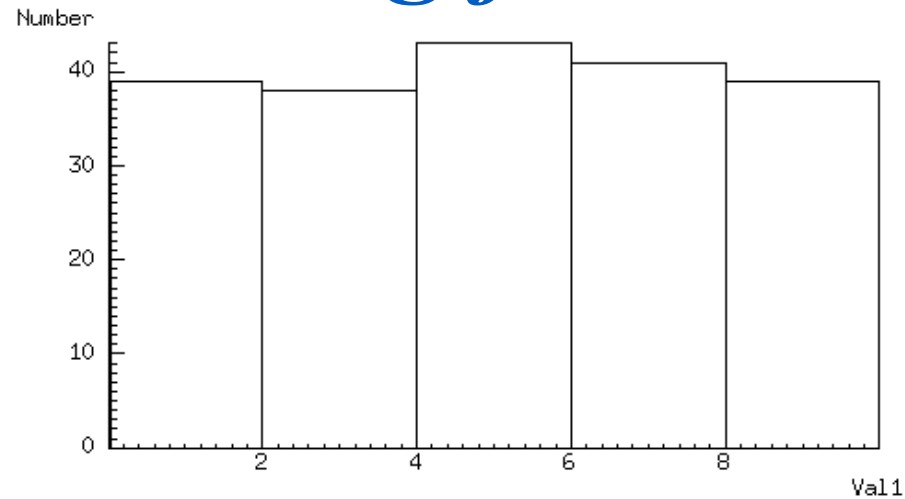
Histograms



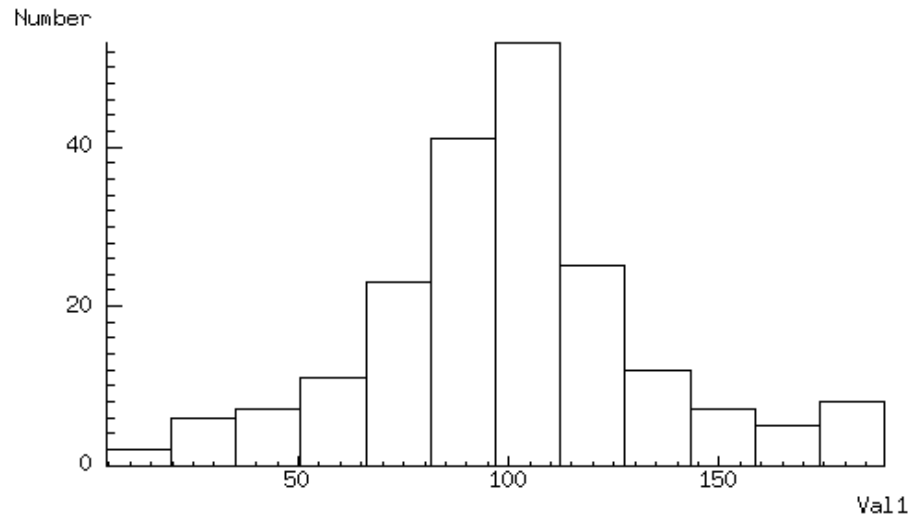
- n Show distribution of values for a single field
- n Bars = "buckets" - width automatic or user-defined
- n Variation in bucket width can reveal/hide details



Interesting features (1)

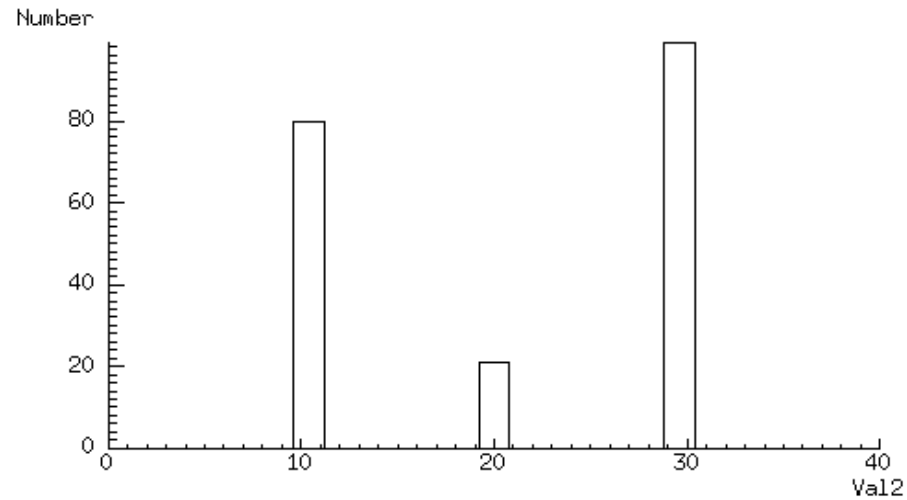


Reasonably even distribution
(larger the data set, more significant the variations)

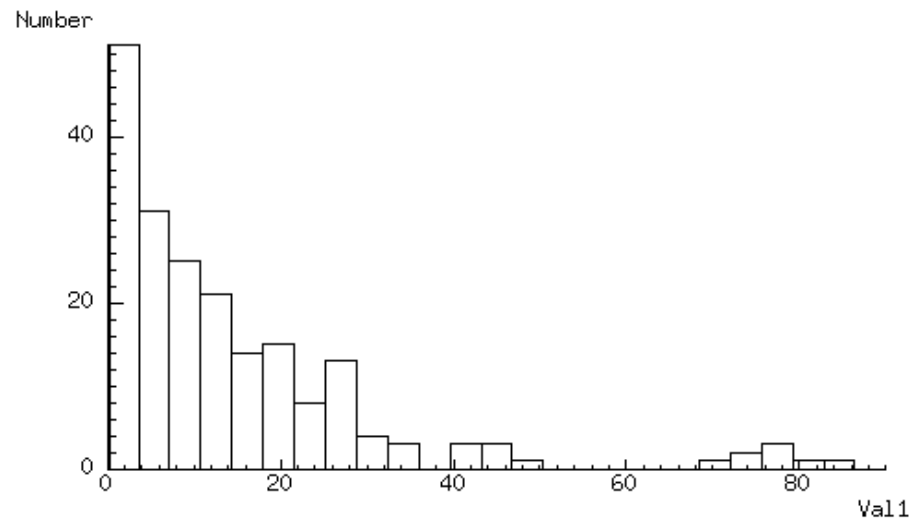


(Approximate) "Normal" distribution

Interesting features (2)



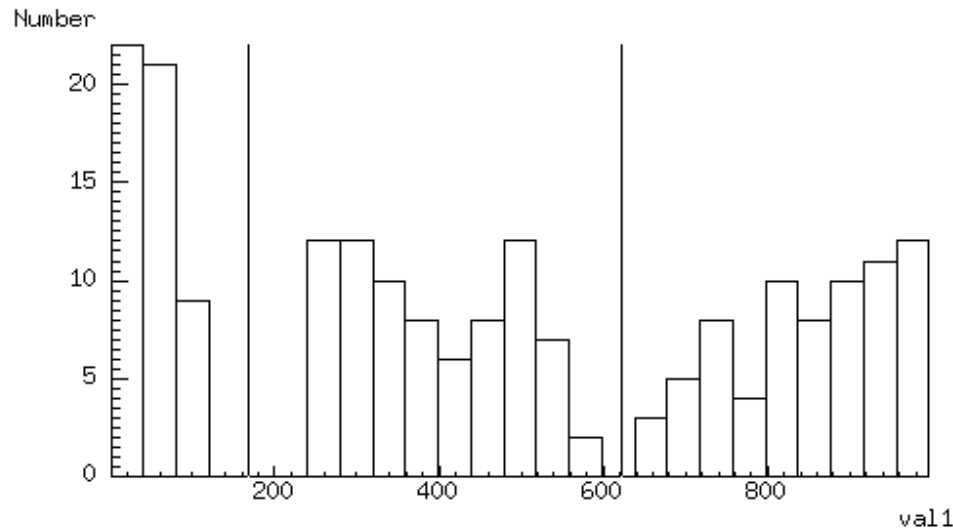
Discrete values - treat as symbol?



Distribution with clustered outliers

Interacting with histograms

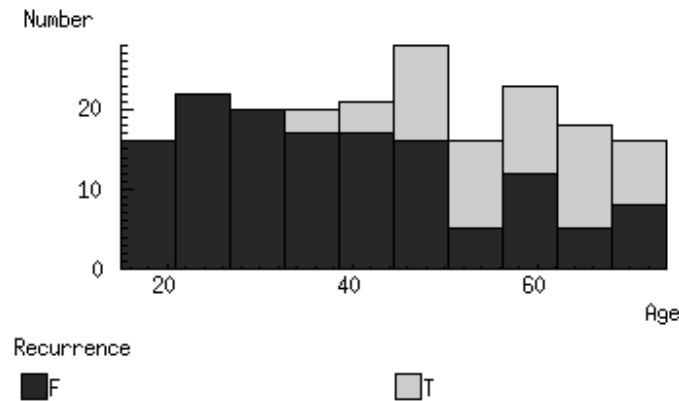
- n Can define bands using the mouse



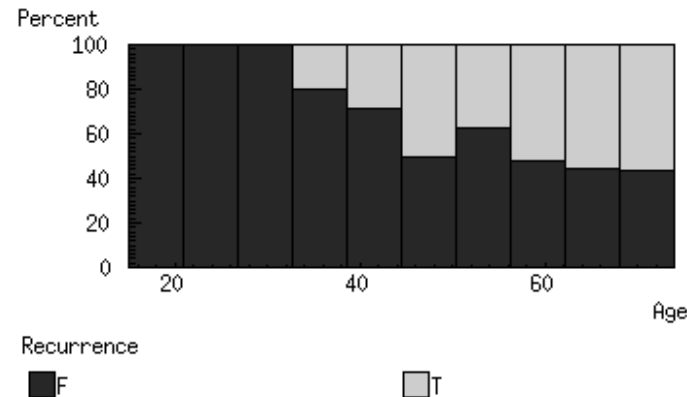
- n Band defined by range e.g. $0.5 < x < 1$, $x > 30$
- n Facilities:
 - Select records on the basis of being in a particular band.
 - Create a flag to indicate whether a record is in a particular band.
 - Add a field which indicates which band a record is in.
- n NB unlike plots, a record can never be in 0 bands.

Normalising Histograms

- n Difference in bar heights can make it difficult to see how proportions vary:



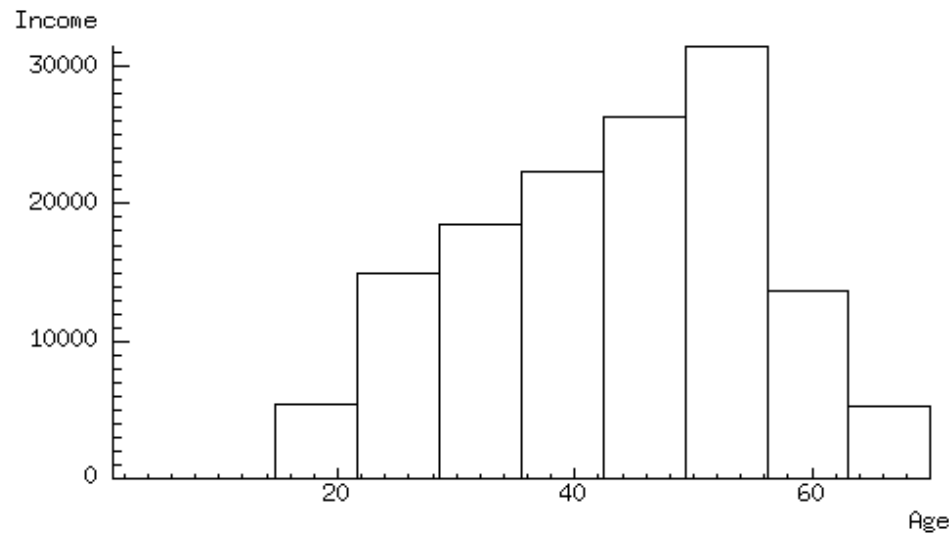
- n Corresponding patterns/features may not be easy to see
- n Normalising scales to percentages, showing proportions:



Collection

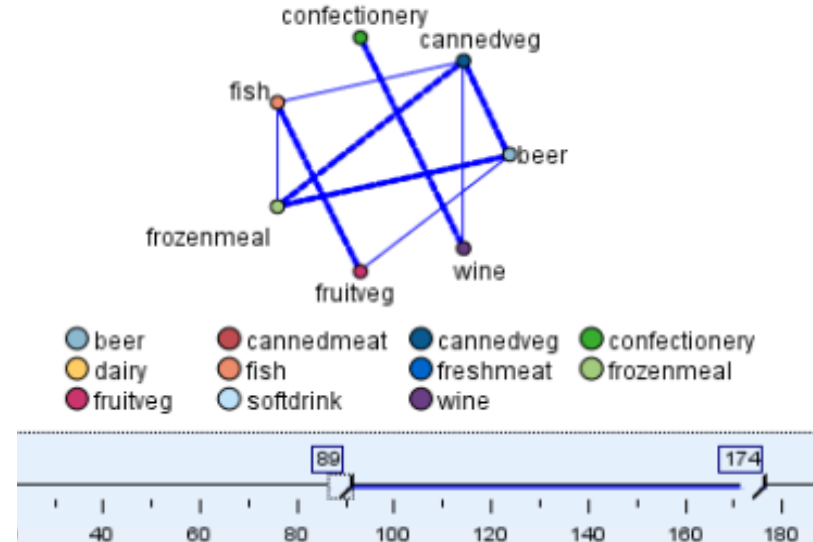


- n Special case of histogram
- n Shows distribution of one variable over values of another
e.g. "Show average income over age"



- n Interactions as for histograms

Webs



n Show strength of connection between symbolic values

"Products 66 & 65 are linked to high income."

"Product 67 is associated with low income and high activity."

"Product 88 is strongly connected to medium income, and to a lesser extent to low income."

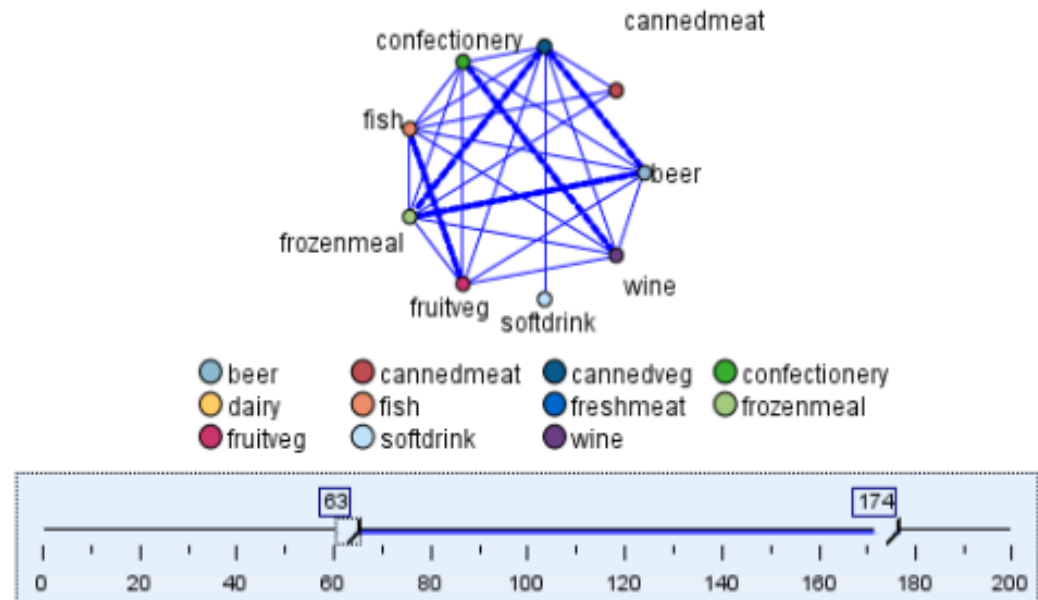
n Weight of line indicates strong/medium/weak connection

n Can specify thresholds for different strengths and minimum strength to display

n Strength can be absolute (number of records) or proportional

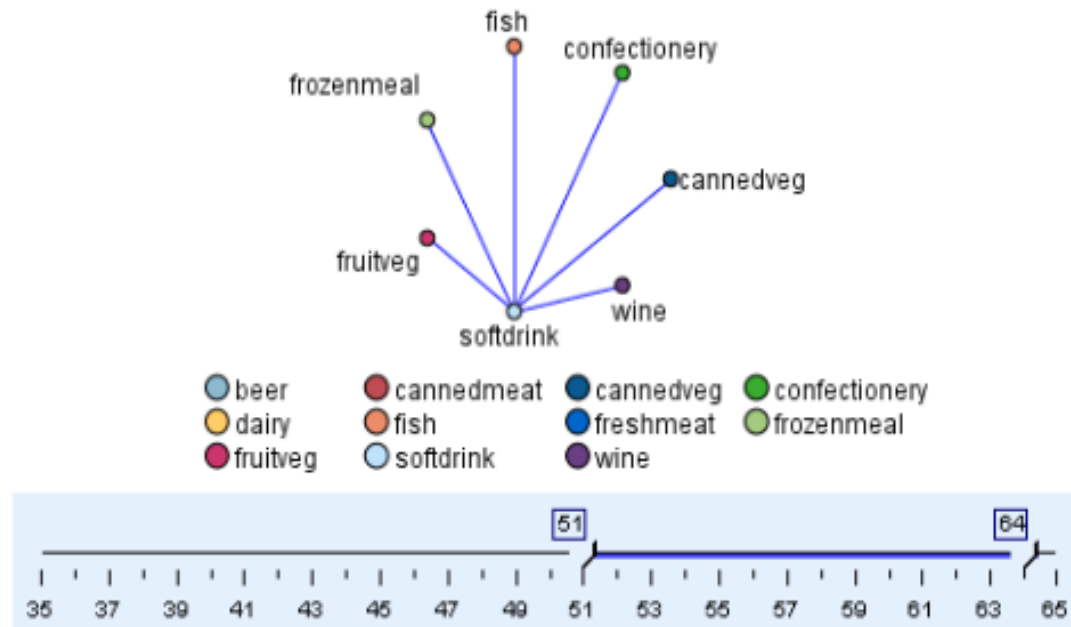
Interacting with Webs

- n Can move or hide points
- n Can select set of lines:

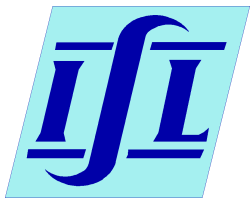


- n For single line or for selection, can:
 - l select records on line(s)
 - l create a flag to indicate a record is on line(s)
- n Can generate textual summary of connections
- n Thresholds can be changed interactively

Directed Web



- n Directed Web is a variant of the Web Node.
- n Shows only connections from any number of 'FROM' fields to a **single** 'TO' field (e.g. drug).



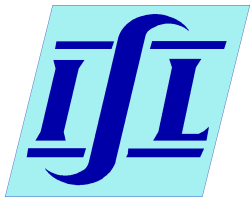
Matrix



- n Provides cross-tabulation of two symbolic fields
- n Cells hold count, or percentage by row/column

Drug					
BP	drugA	drugB	drugC	drugX	drugY
HIGH	23	16	0	0	38
LOW	0	0	16	18	30
NORMAL	0	0	0	36	23

- n A 3rd (numeric) field can be overlaid
- l Sum, Average, Minimum, Maximum
- n Can select cell(s) and generate select/derive nodes based on row/column field values



Statistics

Statistics (1)

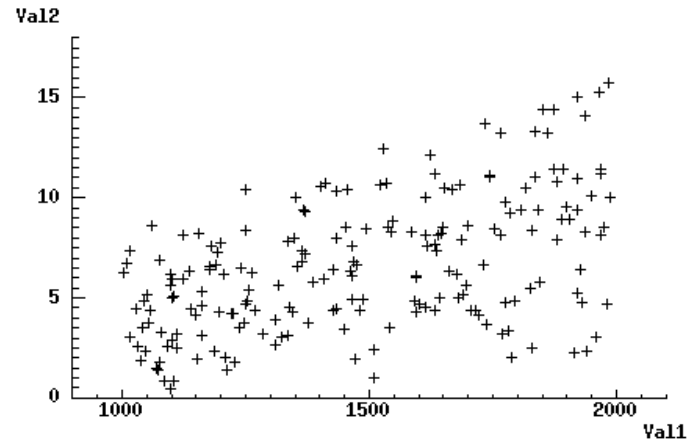


Statistics

- n Analyse single field or relate multiple fields
- n In Clementine, statistics node gives:
 - | Minimum Value
 - | Maximum Value
 - | Number of Occurrences
 - | Mean
 - | Standard deviation
- n Linear correlation with other fields
 - 1 perfect positive correlation
 - 0 no correlation
 - 1 perfect negative correlation

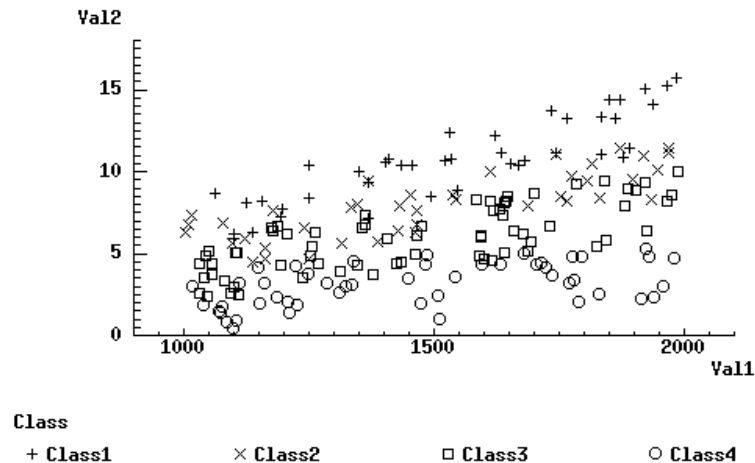
Statistics (2)

Whole population :



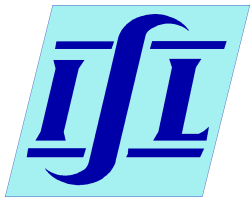
Correlation = 0.510

Overlaying Class field :



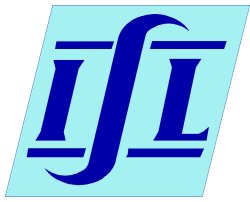
Correlation for individual classes gives:

Class1 : 0.875 Class2 : 0.807 Class3 : 0.739 Class4 : 0.499



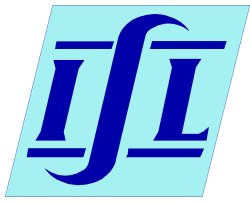
Workshop 2

- n View and interact with the data using
Distribution node, Plot node, Web node
- n Introduce Overlay Fields

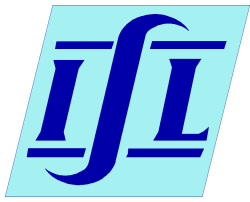


Summary

- n Can display data in many ways
 - l distribution, plot, histogram, collection, web, directed web, matrix
- n Displays are interactive
 - l can generate e.g. select nodes, derive nodes
- n Statistics node can indicate simple relationships
 - l often “clues” for further investigation

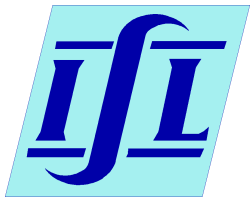


Data Manipulation



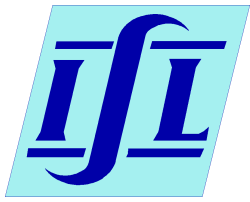
Hypothesis Testing

- n Program own ideas in Clementine/SPSS
Modeller using select and derive nodes
- n Expressions written in CLEM language
- n This section:
 - | CLEM expression language
 - | select & derive nodes
 - | streams for hypothesis testing



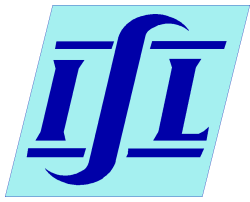
CLEM

- n Rich expression language
- n Includes:
 - Arith/Maths **+, -, /, **, etc.**
 - sqrt, cos, arcsin, exp, etc.**
 - Comparisons **==, >, =<, etc.**
 - String/character
 - Type testing
 - Time/date
 - Sequence handling
- n Any word which is not a function name is assumed to be a field name
- n Field names with spaces or "funny" characters must be 'quoted'



CLEM Values

- n Integers **-999, 0, 1234999999999999**
- n Float **1.234, 1.7e-2, -0.00001**
- n Strings **'fred', 'a piece of text', '999'**
- n Flags **True or false value**
- n In expressions can also use:
 - l character codes (ASCII) **`a` (= 97)**
 - l lists **[10 20 30], ['High' 'Low']**



CLEM Examples (1)

$(V1 + V2 + V3 + V4 + V5)/5$

Average of fields V1...V5

`sqrt (A**2 + B**2)`

Simple math function (Pythagoras?)

`Profit > 0.1 * Revenue`

True if profit greater than 10% of revenue

`ID == 'isl'`

Test for specific string

`issubstring ('Mc', Name)`

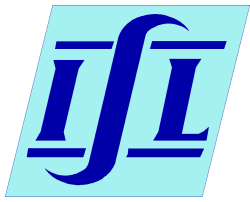
True if Name contains 'Mc'

`strmember (`x`, Code)`

True if code contains the letter `x`

`member (Class, [10 30 70])`

True if Class is 10, 30 or 70



CLEM Examples (2)

date_years_difference (BirthDate, JoiningDate)

Age at joining

time_before (Time1, Time2)

True if Time1 is before Time2

Select Nodes

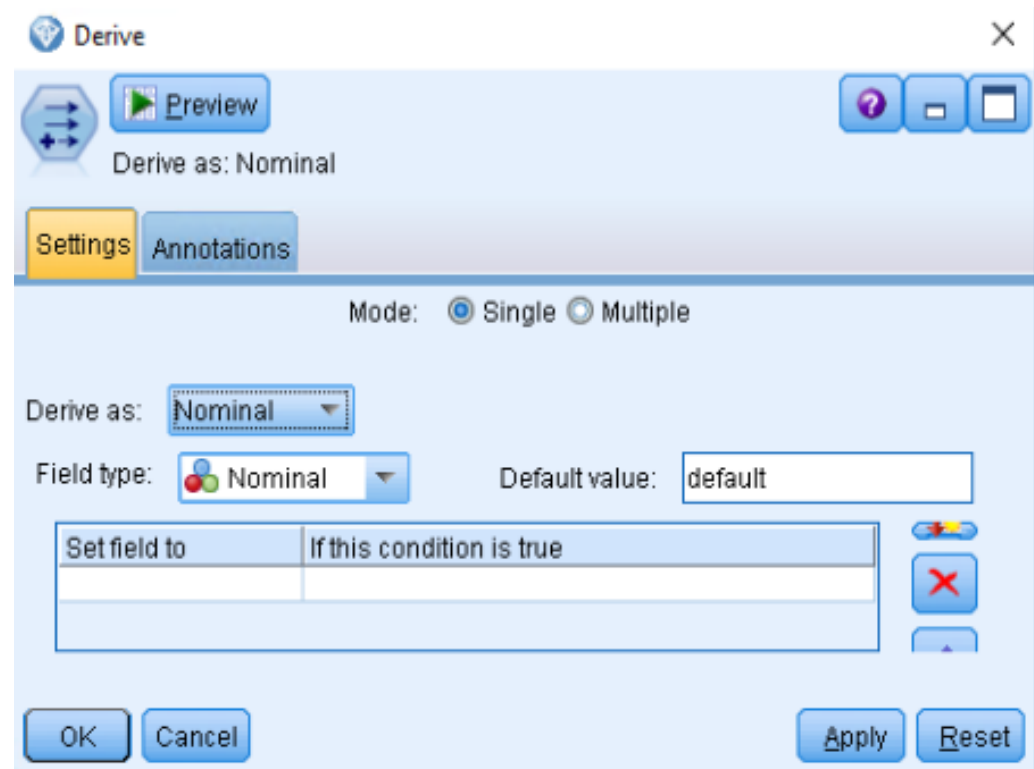


- n Select/discard on specified condition
Value < 0.0
- n Define sub-populations
Sex == 'M' and Age > 50

Derive Nodes



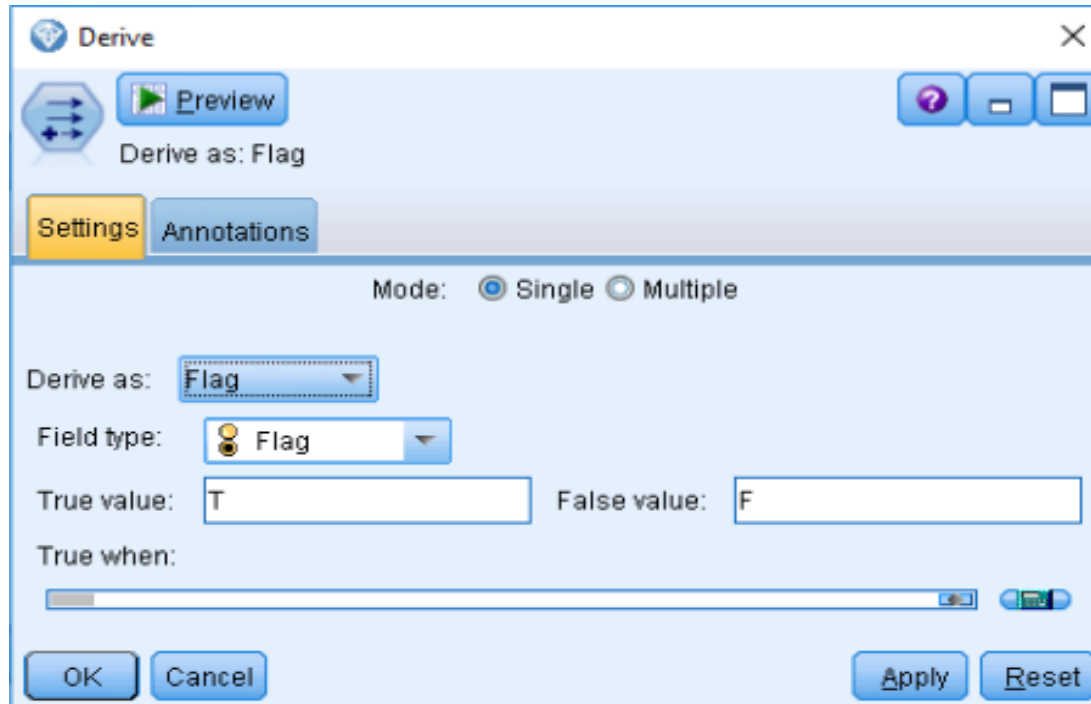
- n Derive nodes create new fields. Several type
- | Any expression
- | Flag
- | Set
- | Conditional



n "Any": <expression> → field value

Derive Nodes - Flag

n "Flag": if <condition>
 then
 <"true" value> field value →
 else
 <"false" value> field value →



Derive

Preview

Derive as: Flag

Settings Annotations

Mode: ☒ Single ☐ Multiple

Derive as: Flag

Field type: Flag

True value: T False value: F

True when:

OK Cancel Apply Reset

Derive Nodes - Set

n

"Set": if <condition1>
then

<value 1> field value



else if <condition2>

then

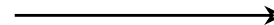
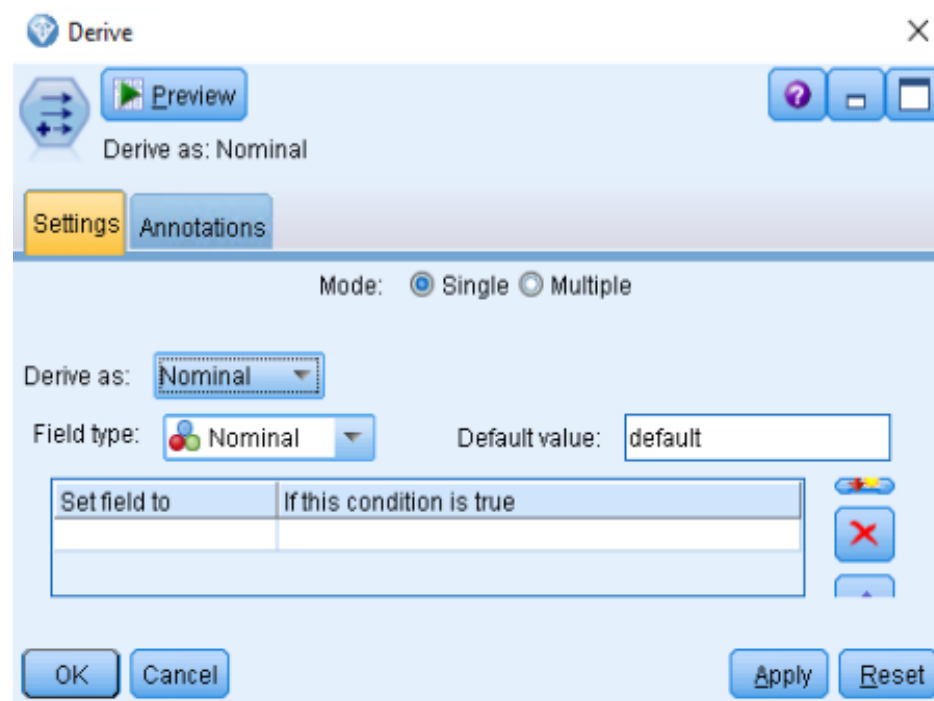
<value 2> field value



.....

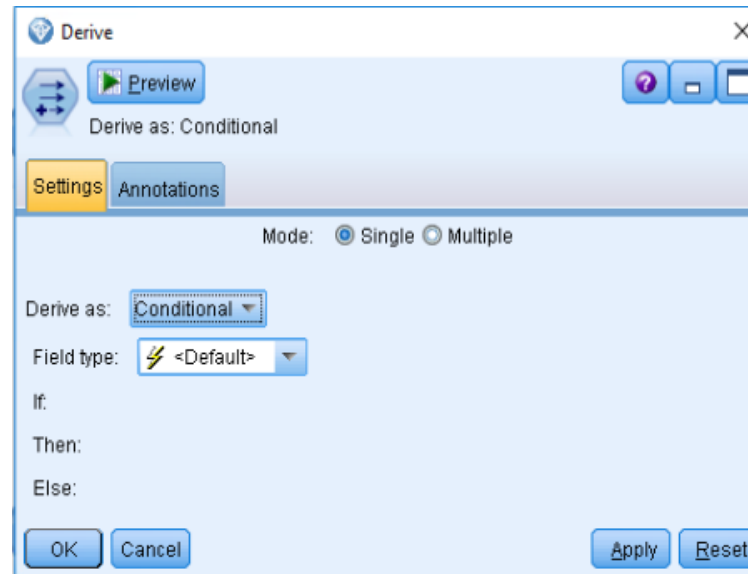
else

<default value> field value

Derive Nodes - Conditional

n "Conditional": if <condition>
 then
 <expression 1> field value →
 else
 <expression 2> field value →



(2 other types for time-series applications)

Hypothesis Testing (1)

e.g. "Profit is usually less than the square root of revenue"

DATA



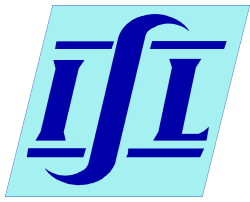
DERIVE: LT_SQRT

Flag: true if : Profit < sqrt (revenue)



DISTRIBUTION: LT_SQRT

Hypothesis			
Distribution			
Value	Proportion	%	Occurrences
F		5.0	10
T		95.0	190



Hypothesis Testing (2)

e.g. "Everyone between 20 and 35 with high blood pressure responds to drug B"

DERIVE: HypDrug

Conditional:

If: Age > 20 and Age < 35 and BP == 'HIGH'

Then: 'drugB'

Else: 'Unknown'

↓
SELECT: HypDrug == 'drugB'

↓
DERIVE: Score

Flag:

True value: Correct

False value: Wrong

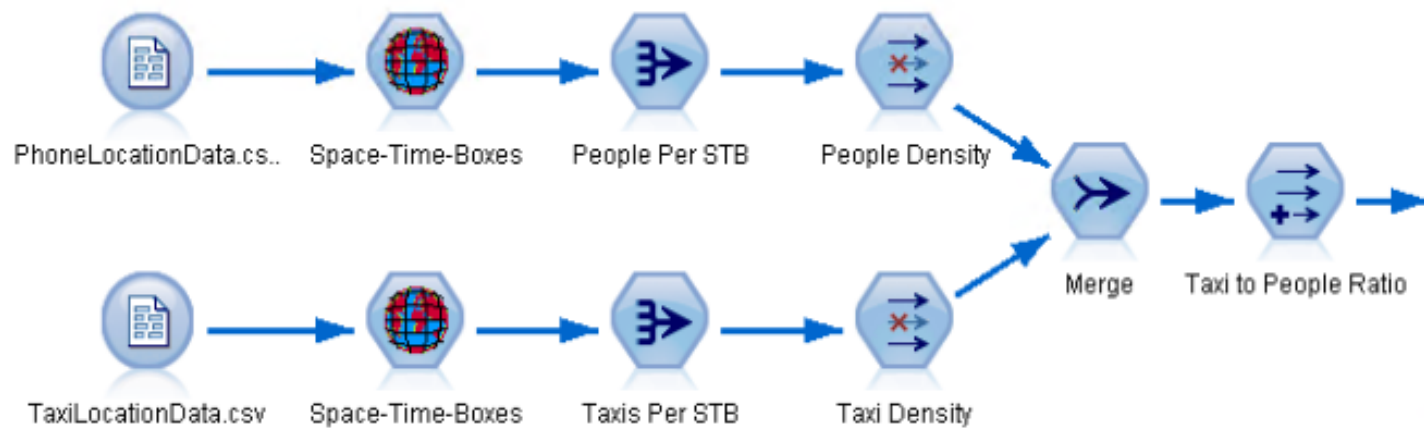
True if: Drug == HypDrug

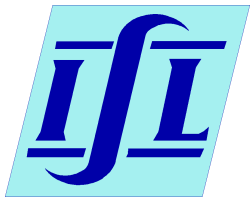
↓
DISTRIBUTION: Score

Merge Nodes



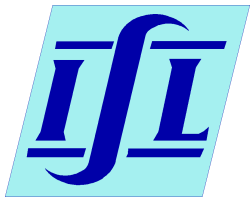
- n Combine records from two or more streams
- n Can combine by:
 - l order - 1st with 1st, 2nd with 2nd, etc.
 - l key - like database "join"





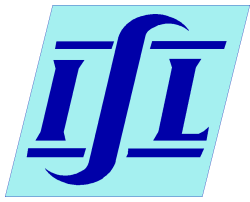
Workshop 3

- n Manipulate the data by deriving new fields
- n View derived fields with histograms
- n Select subsets of the data

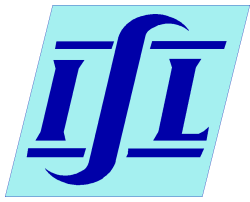


Summary

- n This section has covered 'Passive'
Assistance to Data Mining using
 - | graphical investigation
 - | hypothesis testing
- n It is down to the user to ask questions and follow 'hunches'
- n Modelling algorithms can automate the search for both simple and more complex patterns...

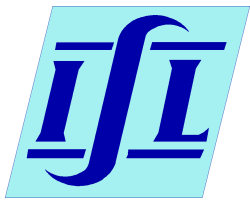


Modelling



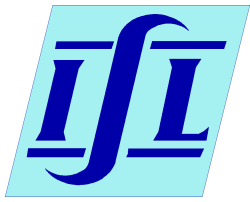
Techniques

- n “Passive” Computer Support
 - | Browsing & Visualisation
 - | Statistics
 - | Hypothesis testing
- n **“Active” Computer Support**
 - | **Neural Networks**
 - | **Rule Induction**
 - | **Techniques combined**
 - | **Regression Modelling**



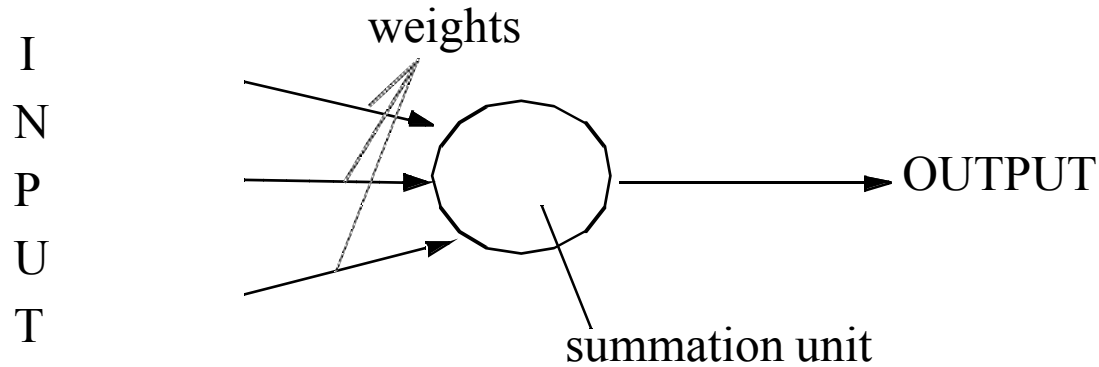
Machine Learning

- n Neural Networks
- n Rule Induction
 - l “Learn” from past cases
 - l Useful for forecasting, prediction, classification, recognition etc.
- n Linear Regression
 - l automatically build numeric models linking numeric attributes
- n Kohonen Networks
 - l Useful for clustering data
- n GRI & Apriori
 - l Useful for association detection

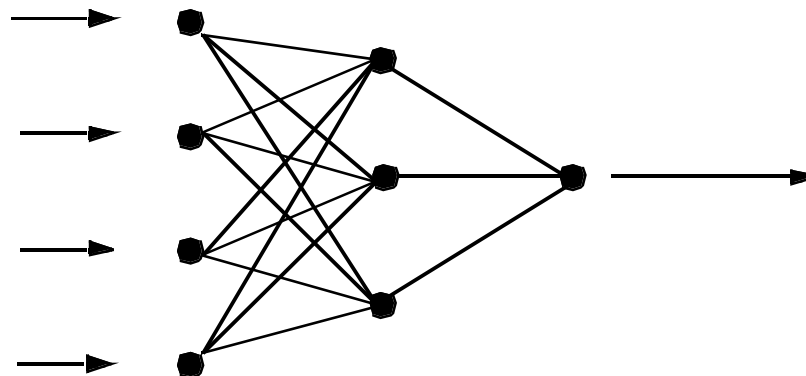


Neural Networks

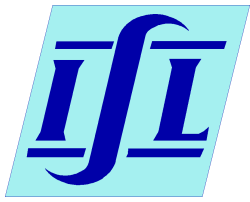
Neural networks (1)



- n Simple model of part of nervous system
- n Basic unit is neuron, a simple summation device
- n Power of networks comes by combining neurons, usually into layers



- n Shape/organization of network = "topology"

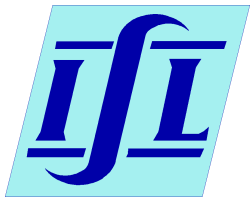


Neural networks (2)

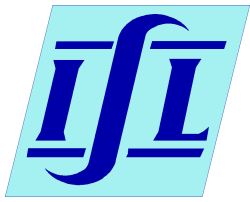
- n Initial predictions are random
- n Learns by feeding the prediction error backwards through the net, in order to modify the weights
- n Particularly strong in numeric applications

Problems

- n Neural networks are powerful, but opaque
- n Internals of trained network simply a set of numbers
 - l very difficult to relate this back to application in any meaningful way
- n Will attempt to use all input variables in simplest mode



Rule Induction

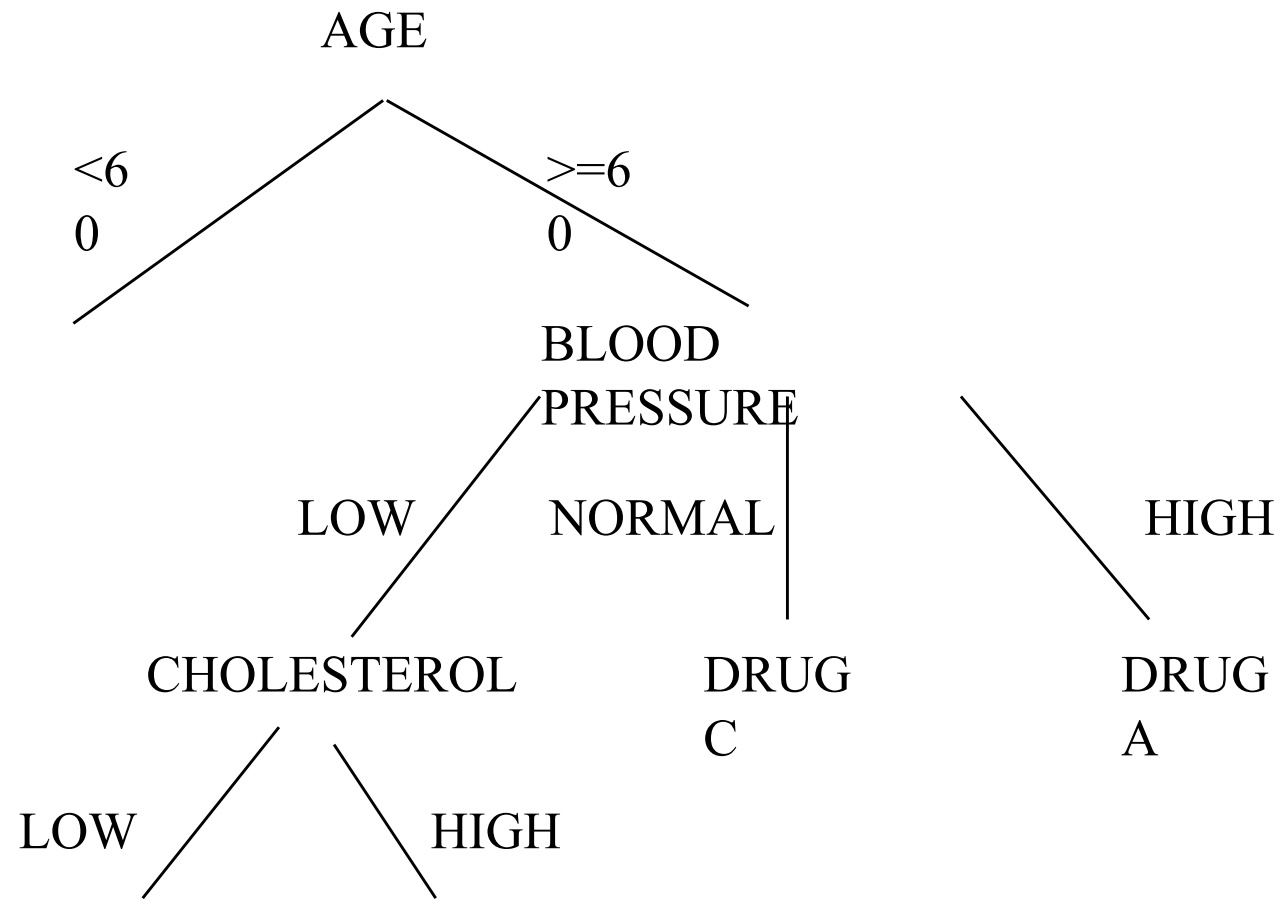


Rule Induction

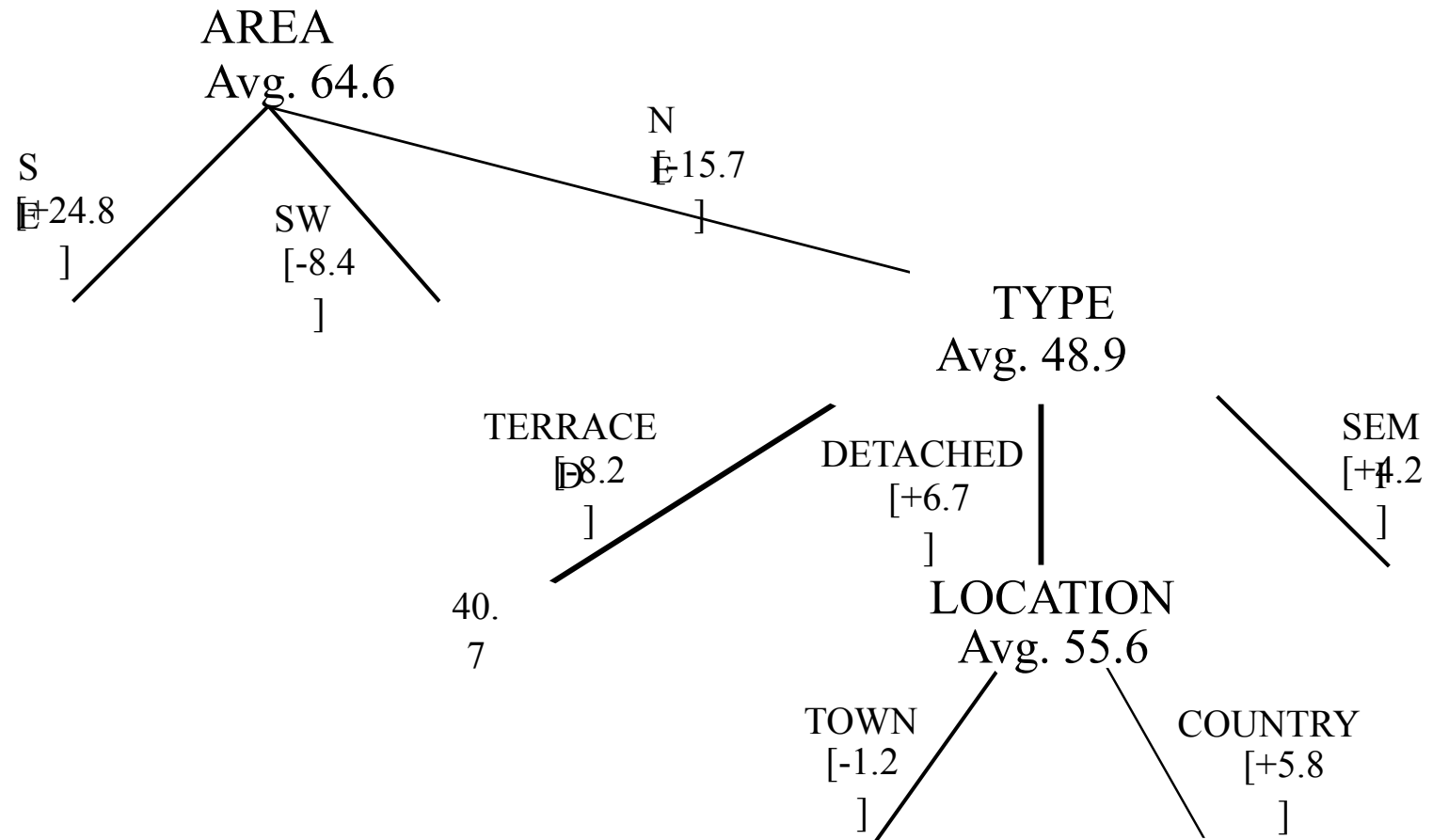
- n Builds "decision tree" representing rule for distinguishing between different cases
- n Uses statistics/information theory to rank fields during tree building
 - optimal use of fields
 - omission of irrelevant fields
- n Particularly strong in symbolic problems. e.g. classification
- n Field's position in tree reflects its importance

Understanding Rule Trees

- n For symbolic outcome



For numeric outcome

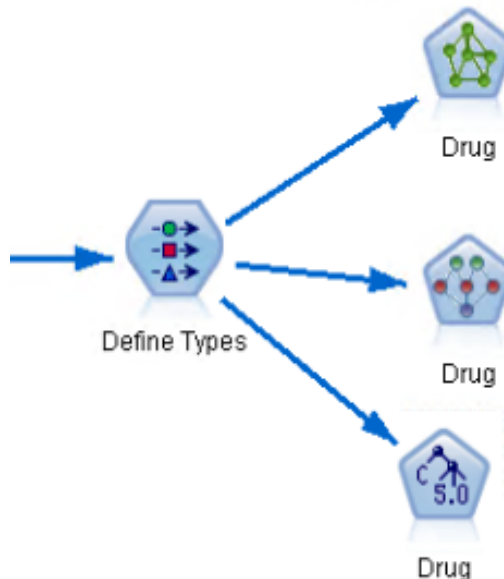


n

"Drop-out" effect for unknown cases

Machine Learning in Clementine/SPSS Modeller

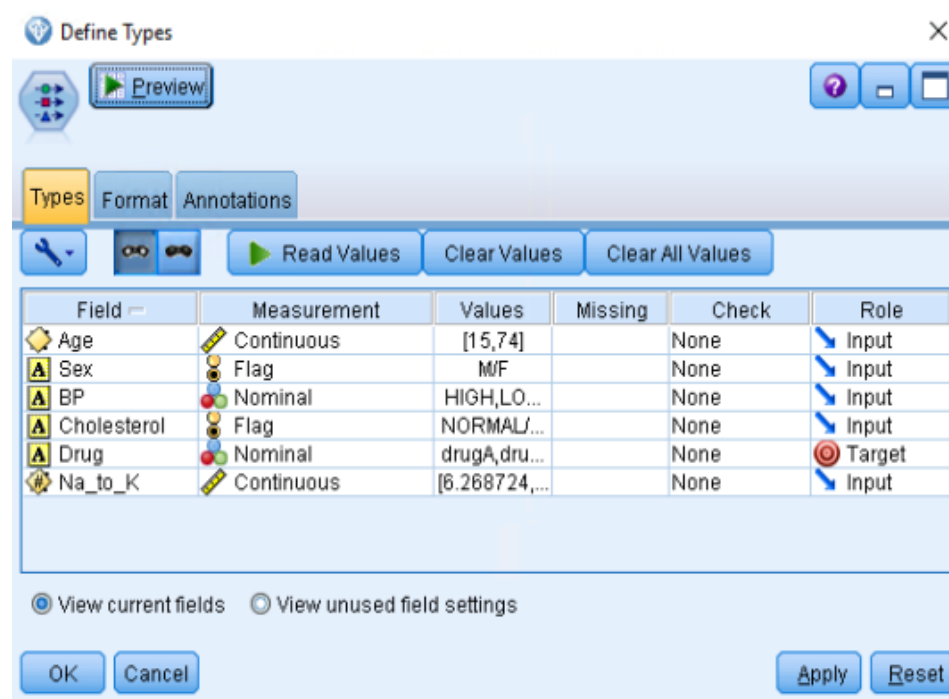
- n Network training and rule induction are terminal nodes
- n First need to go through type node



Type Nodes

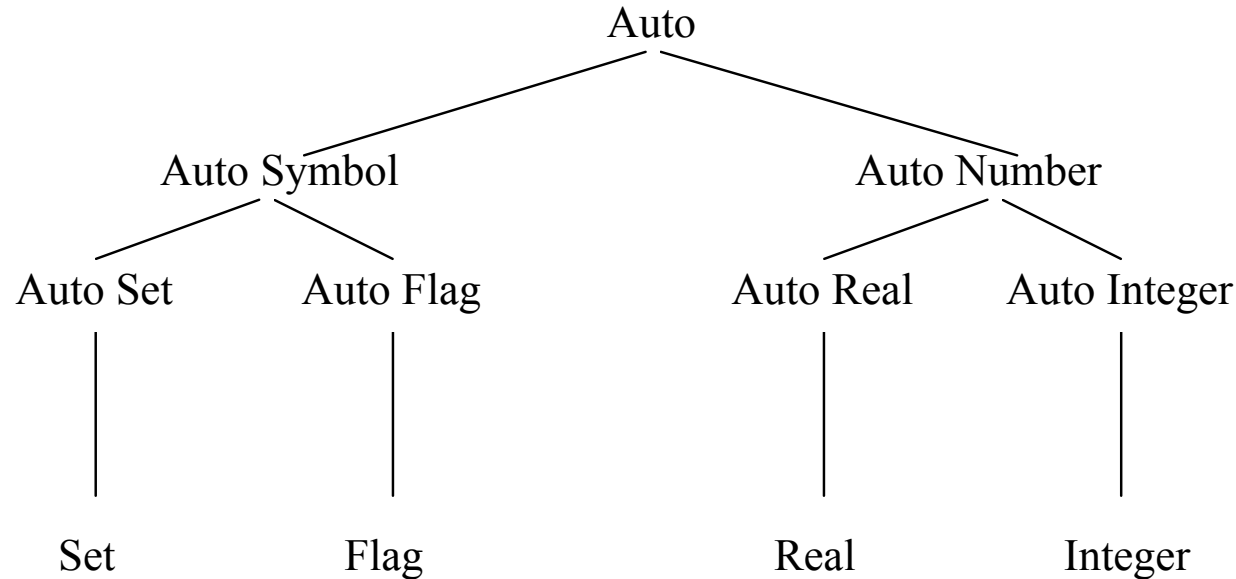


- n Specify type of field, its direction, and blanks
- n Can check for type violations: coerce, warn, discard records, abort



- n Directions can be IN, OUT, BOTH or NONE
 - IN used as input for prediction
 - OUT to be predicted
 - BOTH treats field as both input and an output (for GRI)

Typing in Clementine/SPSS Modeller



- n Auto suitable for most purposes
- n Types are derived from observed values
- n Typically override to:
 - l specify numbers are really symbolic classes
e.g. 1/0 as a flag, 10/20/30 as class labels
 - l broaden range if observed values may be too narrow
e.g. 0.07 - 0.97 -> 0.0 - 1.0
- n Special “Typeless” type

Train Net Node

- n Specify high-level strategy rather than technology-level parameters
- n Training:
 - "Quick": expert system looks at data, chooses network topology and training parameters
 - "Dynamic": starts with simple network, changes parameters and topology in response to training progress
 - "Multiple": creates several (up to 60) networks, trains in parallel
 - "Prune": prunes weak neurons from network
- n Stopping:
 - Default "good", stable solution, or threatened overtraining
 - Specified accuracy
 - Specified amount of training (cycles)
 - Specified time
- n Automatic selection of best network

Build Rule Node

- n Noise: can prevent there being a rule which holds for all cases
- n Accuracy vs. Generality:
 - Accuracy: try to get a rule which fits all cases
 - Generality: build rules from data subsets, prune off subtrees with low coverage



Tree-AS



Random Trees



C&R Tree

C5.0



- n Provides a better rule induction mechanism for symbolic output

- n C5.0 can build Decision Trees or Rule Sets:

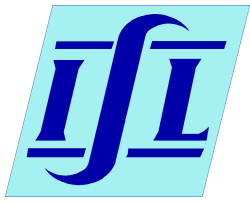
```
if  <condition 1>
and  <condition 2>
and  .....
then <conclusion>
```

e.g.

```
if  Age >= 60
and  BLOOD PRESSURE == HIGH
then  DRUG A
```

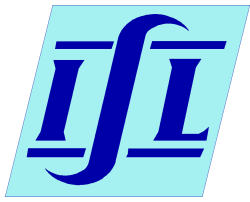
- n In some situations rule sets may be more understandable than large and bushy decision trees

- n C5.0 has comprehensive noise handling ability



C5.0 - Boosting

- n By default C5.0 builds a single classifier (Decision Tree or Ruleset)
- n Boosting instructs C5.0 to build a number of classifiers
- n The model consults each classifier before making a decision
- n Boosting often gives a higher accuracy at the expense of taking longer to train and producing less understandable models



Rule Browser

n

Based on "folding"

Fully folded:

Age <30

Age >=30

Fully unfolded:

Age < 30

lgroup >= 7 -> **High Risk**

lgroup < 7

Conviction T -> **High Risk**

Conviction F

lgroup < 4 -> **Low Risk**

lgroup >= 4 -> **Med Risk**

Age >= 30

Conviction T -> **High Risk**

Conviction F -> **Low Risk**

n

Can fold/unfold both breadthwise and depthwise

n

Can show branch coverage

Age < 30 (478)

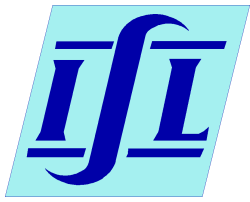
lgroup >= 7 (121) -> **High Risk**

lgroup < 7 (357)

.....

n

Generate option produces filter which cuts down to only the fields used in the rule



Rule Sets

n

Set of "production rules"

Rules for High Risk:

Rule #1 for High risk

if Age < 30

and lgroup >= 7

then -> **High Risk**

Rule #2 for High risk

if Age >= 30

and Conviction == T

then -> **High Risk**

Rule for Medium Risk:

Rule #1 for Medium Risk

if Age < 30

and Conviction == F

and lgroup >= 4

then-> **Medium Risk**

Rules for Low Risk:

Rule #1 for Low Risk

if Age >= 30

and conviction == F

then -> **Low Risk**

Default:

-> **Low Risk** <- catchall

n

Often easier to understand than tree

n

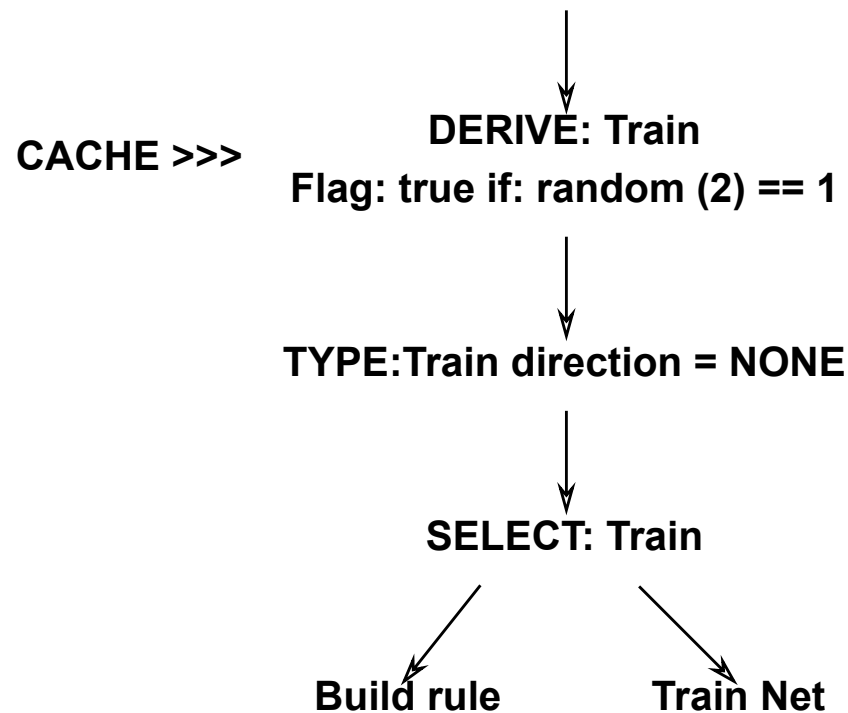
Shows "profiles"

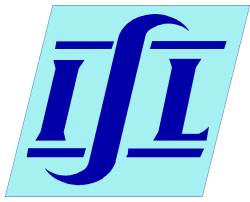
n

As for trees, can show coverage and generate filters

Training & Test Sets

- n Split into training/test sets
- l If want to use same sets in multiple Clementine sessions, write to files
- l For work within the same session, can use tag and cache:





Combining Rules & Nets

Working with Rule & Nets

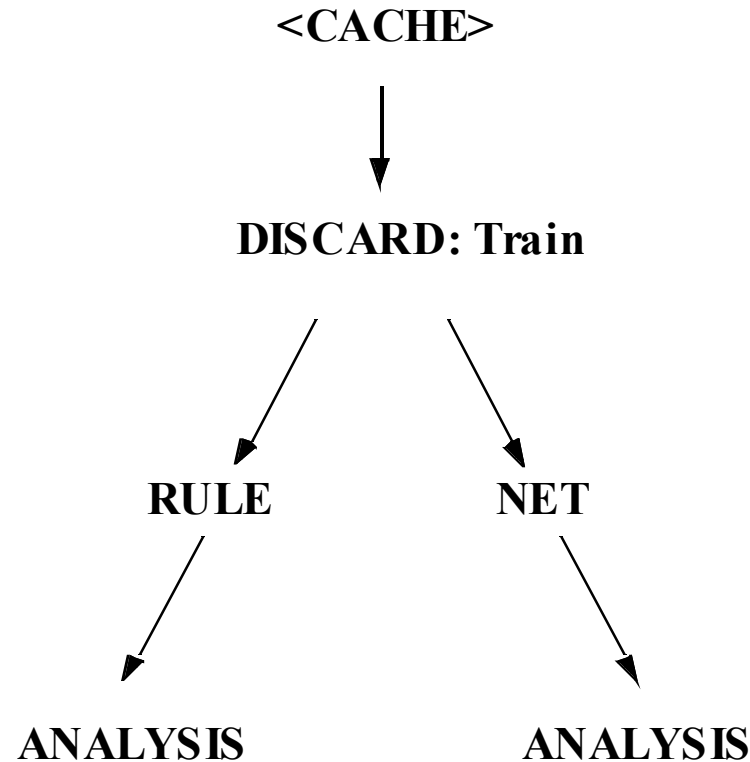
- n Rules & Nets appear on right-hand palette
- n Act as manipulation nodes: add their decisions to record
 - if field was "**profit**", rule's decision is "**\$R-profit**", network's is "**\$N-profit**", C5.0's decision is "**\$C-profit**".
 - "**\$RC-profit**", "**\$NC-profit**" and "**\$CC-profit**" are corresponding certainty figures (0-1).
 - Not available for numeric output
 - can be used like any other fields in expressions, plots etc.

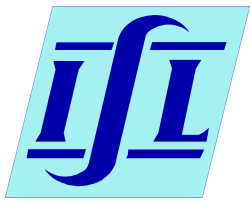


Analysis of Rule & Nets



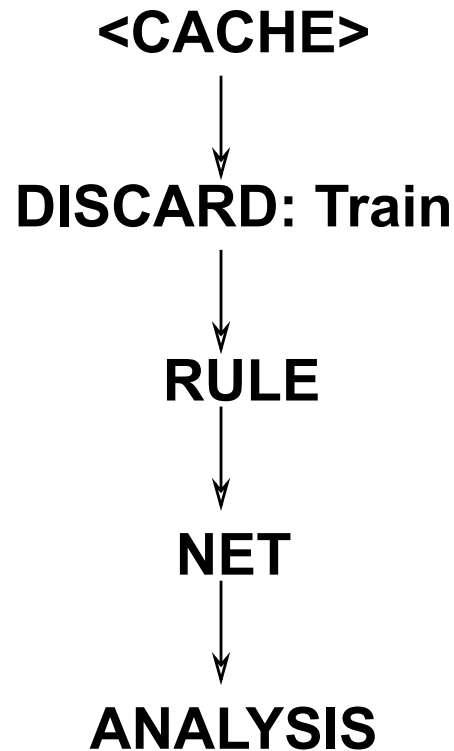
- n Analysis node gives "plotted" assessment of performance:





Combining Rules & Nets

- n Can check each other's performance
- n Can often give improved results



- n For symbolic outcomes, may take **agreement**
- n For numeric, may take **average**

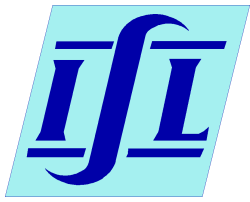
Linear Regression



- n The Regression node builds a multiple linear equation by statistical regression techniques - equation of best fit
- n If input fields are X and Y, and output is Z then the type of equation built is:

$$aX + bY + c = Z$$
 where a and b are the coefficients of X and Y, c is a constant
- n Regression nodes allow complex relationships between numeric fields to be discovered and modelled
- n Result of executing a Regression node is an Equation node
- n Equation Node adds a single new field to the stream:

$$\$E-<\text{outputname}>$$
 E.g.
$$\$E-Z$$
- n Browse or Edit equation node to see the equation

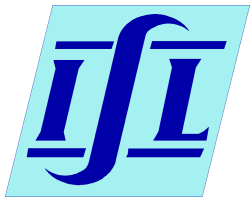


Workshop 4

n Drugs practical

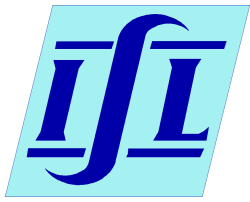
n Aim:

- | Understand how to build models using Rule Induction and Neural Network Techniques
- | Predict which patient should be given which drug
- | Analyse the performance of the models
- | Browse rule based model

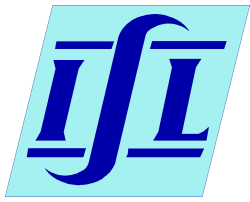


Summary

- n Multiple techniques for building predictive models
 - | neural nets
 - | rule induction
 - | regression
- n Type nodes set up data for modelling
- n Combining multiple models is often beneficial



“Unsupervised” Modelling

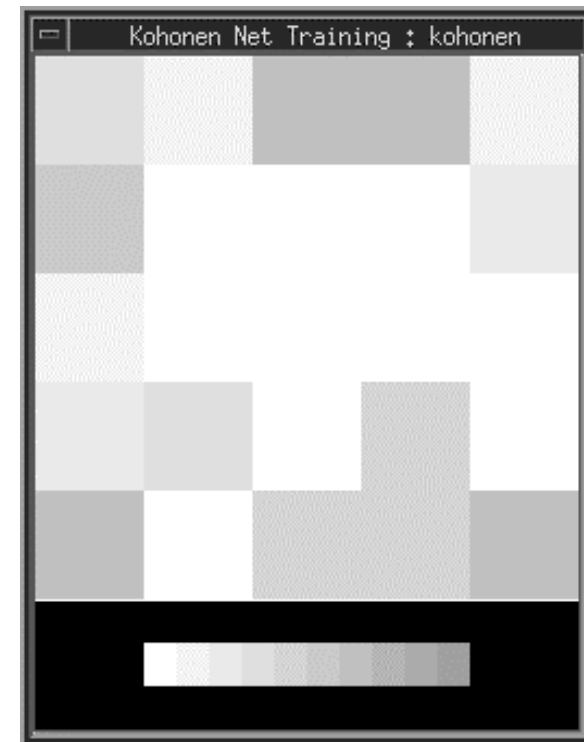


Clustering with Kohonen Networks

Kohonen Networks (1)



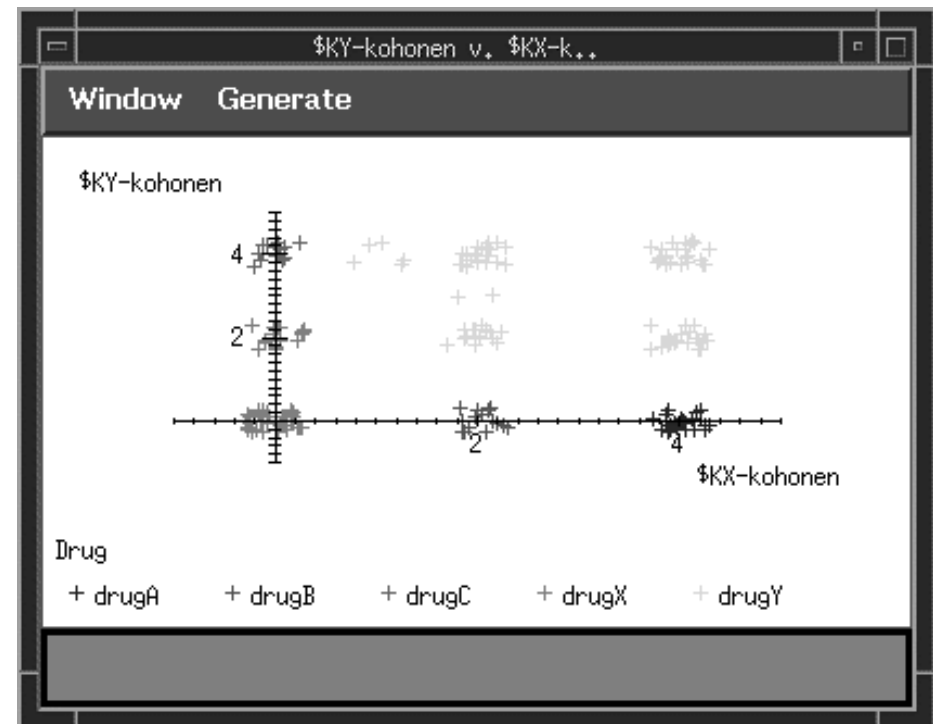
- n Different type of Neural Network
- n Kohonen Networks cluster the data
- n Unlike neural nets and other models mentioned so far
 - knets are 'unsupervised' - they do not require an output field
- n Each neuron in the network competes with its neighbours to win records
- n Similar records appear close in final map
- n Depth of colour represents 'strength' of node



Kohonen Networks (2)



- n Once a Kohonen Network has been trained - it can be used just like any other generated model
- n Output for each record is of form
\$KX-<name>
\$KY-<name>
- n Plot these co-ordinates and agitate the values - overlay symbolic attributes



Kohonen Networks (3)

- n Kohonen Networks do not describe 'why' records have been clustered together
- n Regions on the output plot can be grouped by creating 'derive nodes'
- n These regions can be input to C5.0 to find a description for that region

For example:

Rules for T:

Rule #1 for T:

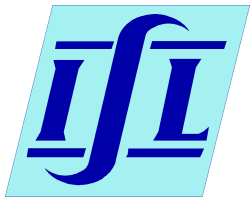
**If Sex == M
and BP == LOW
and Cholesterol == NORMAL
then -> T (19, 0.93)**

Rules for F:

Rule #1 for F:

**if Cholesterol == HIGH
then -> F (103, 0.99)**

Default : -> F

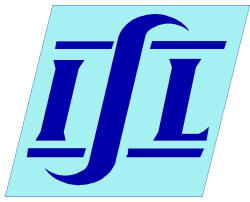


Workshop 5

n Clustering Analysis

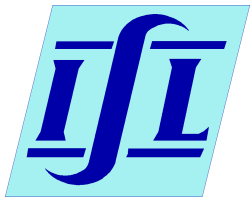
n Aim:

- | Understand how to use Clustering techniques.
- | Use Kohonen Networks to cluster patients.
- | Use Rule Induction to build descriptions of clusters.
- | Find clusters of 'high risk' patients (e.g. high BP and cholesterol level).



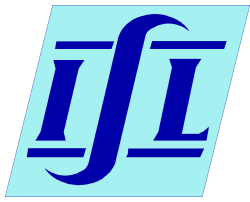
Summary

- n Kohonen networks find clusters of similar records
- n Can be combine with rule induction to provide descriptions (“profiles”, “signatures”) of clusters



Association Detection

- Apriori and GRI



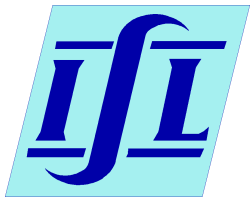
Association Rules

- n Association rules show which things (events, attributes, purchases, ...) typically occur together
- n In Clementine, presented as:
consequent $\leq$ *antecedent1* & *antecedent2* & & *antecedentN*

e.g. **beer** $\leq$ **snacks & newspaper**

“Customers who buy snacks and a newspaper are also likely to buy beer”

- n Consequent sometimes referred to as **head**, list of antecedents as **tail**
- n ***N.B. Association rules do not necessarily indicate a causal relationship - just that certain things occur together***



Coverage and Accuracy

- n Indicate *strength* or *quality* of rule

consequent $\leq$ *antecedent1* & (coverage, accuracy)

- n Coverage (or *Support*):
 - l how often antecedents occur together

- n Accuracy (or *Confidence*):
 - l how often, when the antecedents occur, the consequent also occurs

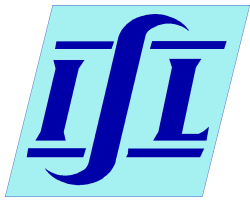
e.g. **beer $\leq$ snacks & newspaper (2744:12.2%, 0.68)**

“12.2% of customers (2744 people in this data set) buy snacks and a newspaper; 68% of such customers also buy beer”

Apriori



- n Efficient association detection algorithm
 - n Only uses symbolic fields
 - | Not sensitive to direction (IN, OUT, BOTH)
 - | Regardless of direction, fields can all appear on both sides of the rule
 - n User controls:
 - | Minimum support and confidence
 - | Maximum number of antecedents
 - | Flag fields used as simple flags:
 - <= previously_cautioned &
- Or sets:
- <= previously_cautioned=T &
 - <= & previously_cautioned=F &...



GRI

- n “Generalised Rule Induction” algorithm
- n Searches for most “interesting” independent rules
- n Fields with direction **IN** appear in antecedents only, **OUT** in consequents only, **BOTH** in either
- n Numeric fields with direction **IN** may be used (appear in antecedents)
- n User controls:
 - l Maximum number of rules
 - l Maximum number of antecedents
 - l Flag fields used as simple flags or sets
 - l Inclusion of “negative” rules
- u = rules which are interesting because they occur rarely

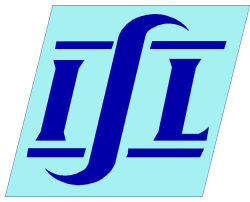
Association Rule Sets

- n The results of Apriori and GRI are represented on the Generated Models Palette as “rough diamond” icons



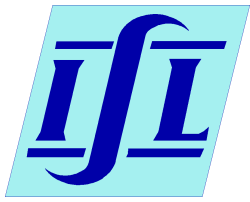
Buying Rules

- n These may not be placed on the stream panel, and may not be exported as C code
 - l i.e. they are not directly executable models
- n They can be browsed in the same way as other models



Browsing Association Rule Sets

- n Use standard Clementine rule browser, with modified options
- n Can order rules:
 - l alphabetically by head
 - l by coverage
 - l by confidence
 - l by product of coverage x confidence
- n Can generate executable rule set to predict specified field
 - l specify minimum confidence/coverage, default value

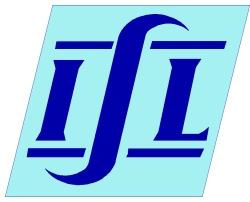


Workshop 6

Association Detection

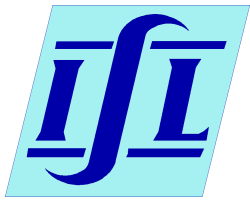
Aim:

- Understand how to use Association Detection
- Use GRI to analyse basket items
- Confirm findings using a Web
- Profile customer groups using Rule Induction
- Use GRI to analyse basket items together with a profile of the shopper

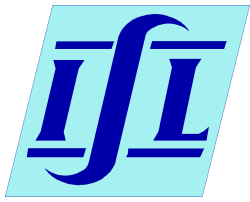


Summary

- n Association rules show things which occur together
 - l not necessarily causal relationships
- n Apriori and GRI both find association rules
- n GRI can use numeric fields in “condition” part of rule

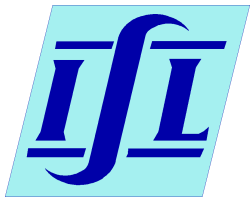


Data Mining Projects



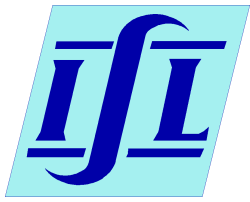
Data Mining Projects

- n Data mining is a process
- n It is not simply applying algorithms and tools to data
- n Business issues are as important as technical ones



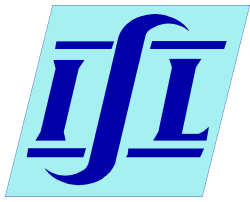
Identifying Potential DM Projects

- n Data available and consistent
- n Room for improvement in current performance
- n Genuine business/operational need
- n Data expertise available
- n [For pilot: moderate scope]



Problem Assessment

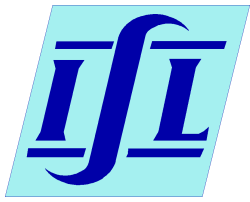
- n Is data available?
- n Does it cover the relevant factors?
- n Is there enough of it?
- n Is the data noisy?
- n Is there expertise available on it?



CRISP-DM:

Cross-Industry Standard Process for Data Mining

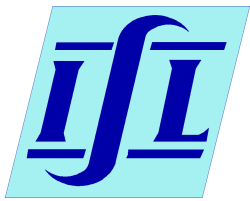
- n Aim: to create a standard, non-proprietary process model (or methodology) for data mining
- n Collaboration between ISL, Daimler-Benz, NCR, OHRA, partially funded by EC
- n Involves Special Interest Group (SIG) of suppliers and large-scales users



CRISP-DM Process Model

- n Decomposes data mining project into *phases* and *tasks*
- n Encompasses business issues
- n Recognises that process is non-linear, iterative

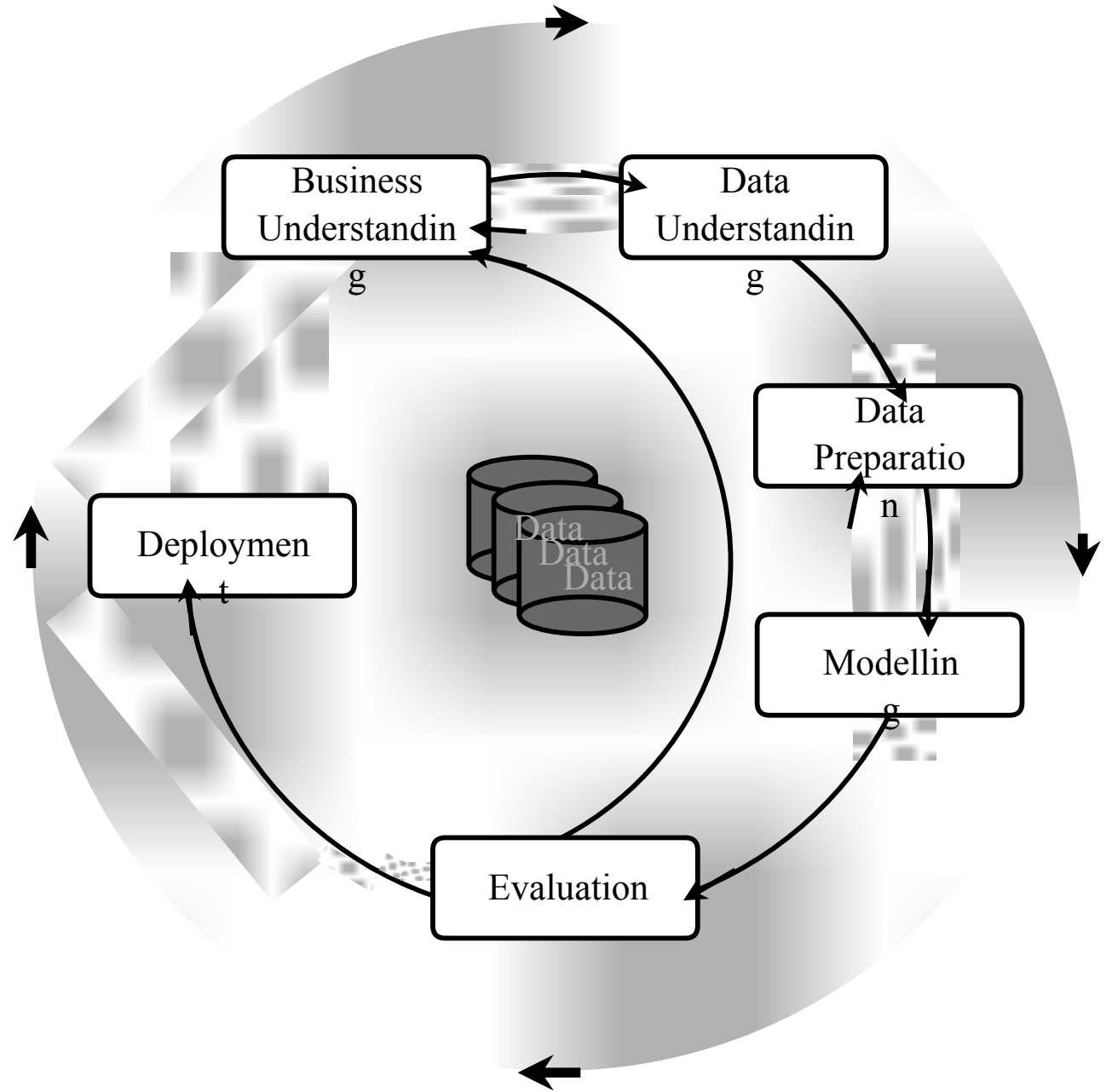
- n First draft in trials from February 1998, final version by end '98
- n ISL implementing first support tool - *CRUST* - on Clementine

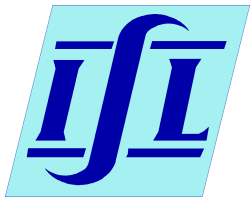


The initial CRISP-DM Process Model: Generic Tasks

Business Understanding	Data Understanding	Data Preparation	Modelling		Evaluation	Deployment
g	g	n	Data Exploration n	Build Model		t
initiate data mining understand terminology specify business - questions - requirements and constraints analyse environment specify data mining goals select tools develop solutions assess costs and benefits identify risks summarise assumptions review success criteria evaluate solutions initial process plan	analyse meta data analyse attribute - domains - relevance - correlations check data quality acquire relevant knowledge validate assumptions review process plan	select data transform data clean data construct data integrate sources review process plan	visualize data form hypotheses validate hypotheses review process plan	review techniques apply techniques evaluate models manipulate data tune parameters review process plan	evaluate results evaluate process review process plan	plan - deployment - monitoring generate report integrate results implement process transfer knowledge monitor results and processes get feedback

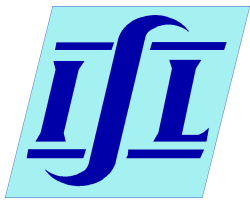
Possible iterations in *CRISP-DM*



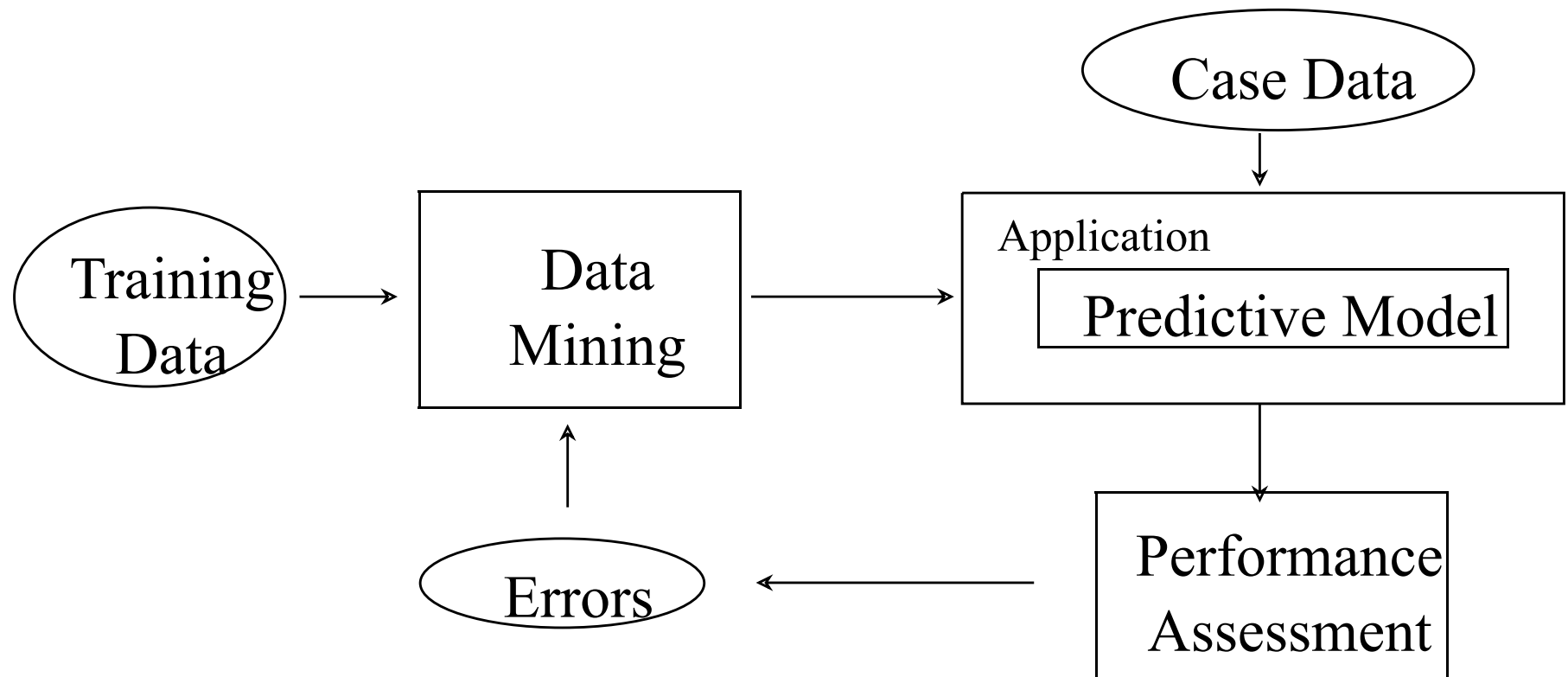


Deployment of DM Solutions

- n Many people choose a DM approach as protection against volatility
- n As situation changes, human knowledge and machine-generated models become out-of-date
 - l performance degrades
- n Models built through data mining can be updated as new data becomes available

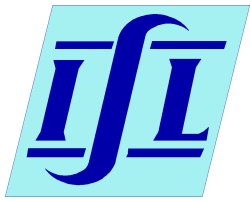


Deployment of DM Solution in a volatile environment



Re-training triggered by:

volume of errors, proportion of errors, fixed time period...

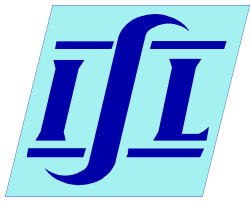


Workshop 7

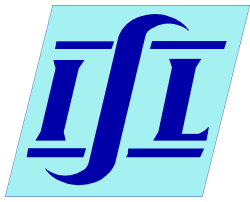
- n Discuss Data Mining Applications

- n Aim:

- | Identify 'pros' and 'cons' for data mining applications
- | Recruitment - application vetting
- | Catalogue Shopping - predict revenue
- | Retail - joint promotions
- | Elections - predict results

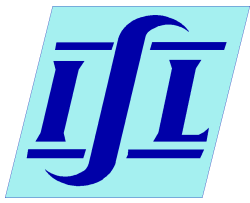


Strategy, Tips & Hints



Data Samples

- n For speed, test initial ideas by experimenting on samples of data
- n Sample Node
- n Regular and random samples
- n Caches "freeze" random samples
- n Make use of Quality Node to enable sample preparation



Attribute Selection

Irrelevant attributes

- n Two extremes:
 - l unique values e.g. record IDs
~ but look out for hidden information
e.g. account number may indicate date
 - l no variation (single value)
- n Ruled out by common sense or domain experts (but sometimes still worth a try)
- n Information subsumed into other (possibly derived) attributes
 - But be careful:
Duration = Finish - Start
but Start and/or Finish times may be important in their own right

Attribute Selection

- n Don't include input attributes which are really outputs.
e.g we have revenue Before and After a Promotion,
and want to predict Increase (= After - Before)
- vital to exclude After
- n This is a special case of using "illegal" attributes:
 - | may be available in historical data, but wouldn't be available at point of prediction
 - | e.g "Audience Appreciation" in Audience Share Prediction
 - | Common cause is lookahead in time series problems

Derived Attributes

- n Data exploration (browsing, visualisation, statistics, hypothesis testing, etc.)
- n Aggregation & Simplification

ZERO	ZERO	ZERO
MINOR	PARTIAL	NON-ZERO
MODERATE	EXTENSIVE	
CONSIDERABLE		
EXTENSIVE		

(look at class distribution)

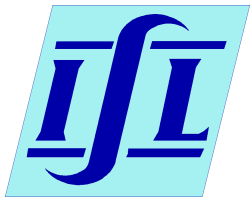
Set ('Low'/'Medium'/'High')

Numeric Range

Flag (Above/below threshold or within specified band)

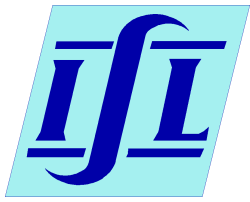
(examining with histograms etc.)

- n Using symbols will favour induction



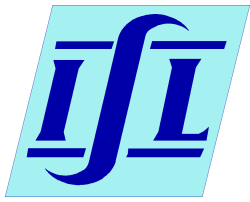
Induction as a guide to Attribute Selection

- n Induction builds rules using significant attributes, ignores others
- n Can use its choice of attributes as a guide



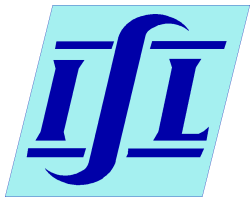
Attributes with Missing Values

- n "GAPS" or "BLANKS" in the data
Values not collected or mistyped
- n The lack of data item can be information itself
- n For symbolic attributes (sets, flags) can simply include these blanks
Problem : Increases complexity of data set.
- | Exclude fields with missing values
Use a filter or type node
- | Exclude records with missing values.
Use a select or type node
- | Insert sensible default values
Using a type and/or filler node
- | Predict missing values (nets, rules)
Train models to predict field values
Insert predictions using filler node



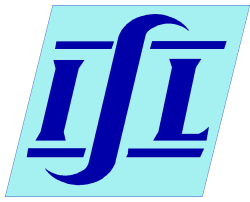
Attributes as Numbers or Symbols

- n Fields may be numbers, but these are really class labels
e.g. `10', `20', `40', `99'
- n Typically give "lumpy" distribution



Unbalanced Data

- n If data contains significant imbalance in outcomes, this can make it difficult to build accurate predictive models using machine learning
- n Most likely to affect:
 - neural net
 - rule induced for generality



Classification Problems

- n Suppose 95% of the data is Class A, 5% Class B; a network only has to return A all the time to get 95% accuracy!
- n Solution 1:
If there is enough data - use a Balance node to 'reduce' the occurrences of the dominant classes
- n Solution 2:
If data is sparse - use a balance node to duplicate the occurrences of the smaller classes

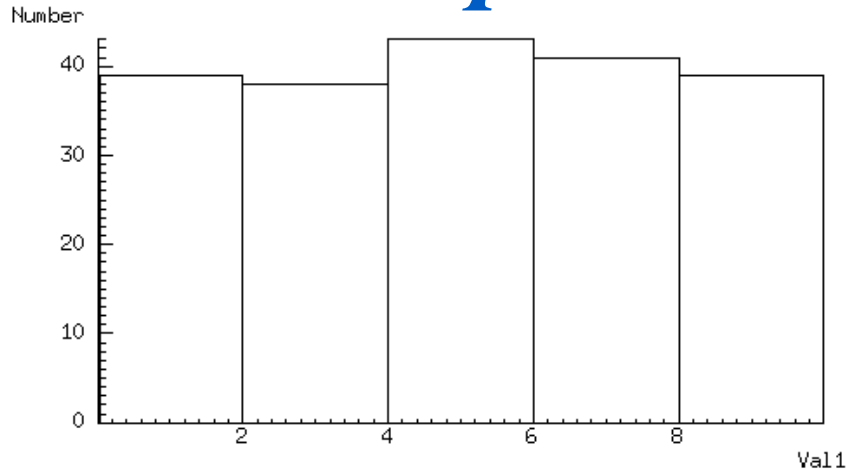
Balance nodes can be:

1. edited manually - the user specifies the boost/reduce factors

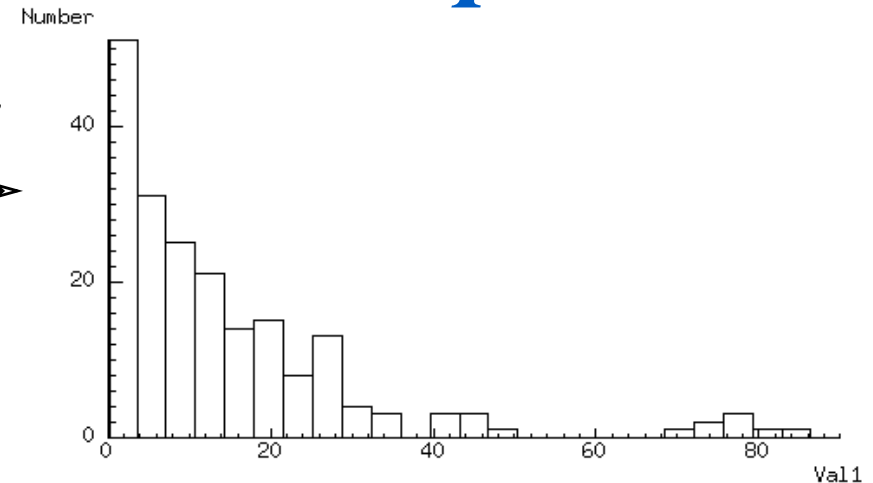
OR

2. created automatically from Distribution displays and Histograms

Numeric prediction/estimation problems



better
than
→



- n Even distribution of values is best
- n Can apply balance node, but excessive for extreme cases.
- n Solution: Apply normalization function

NORMALIZATION

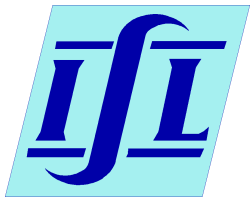
DATA → NORMALIZED TRAIN →
DATA NET

NORMALIZATION

DENORMALIZATION

DATA → NORMALIZED → APPLY → NORMALIZED → RESULTS
DATA NET RESULT → → →

e.g. Normalize Inverse
 $\text{Norm} = \log_{10}(x)$ $x = 10^{\text{Norm}}$
 $\text{Norm} = x^{0.1}$ $x = \text{Norm}^{10}$

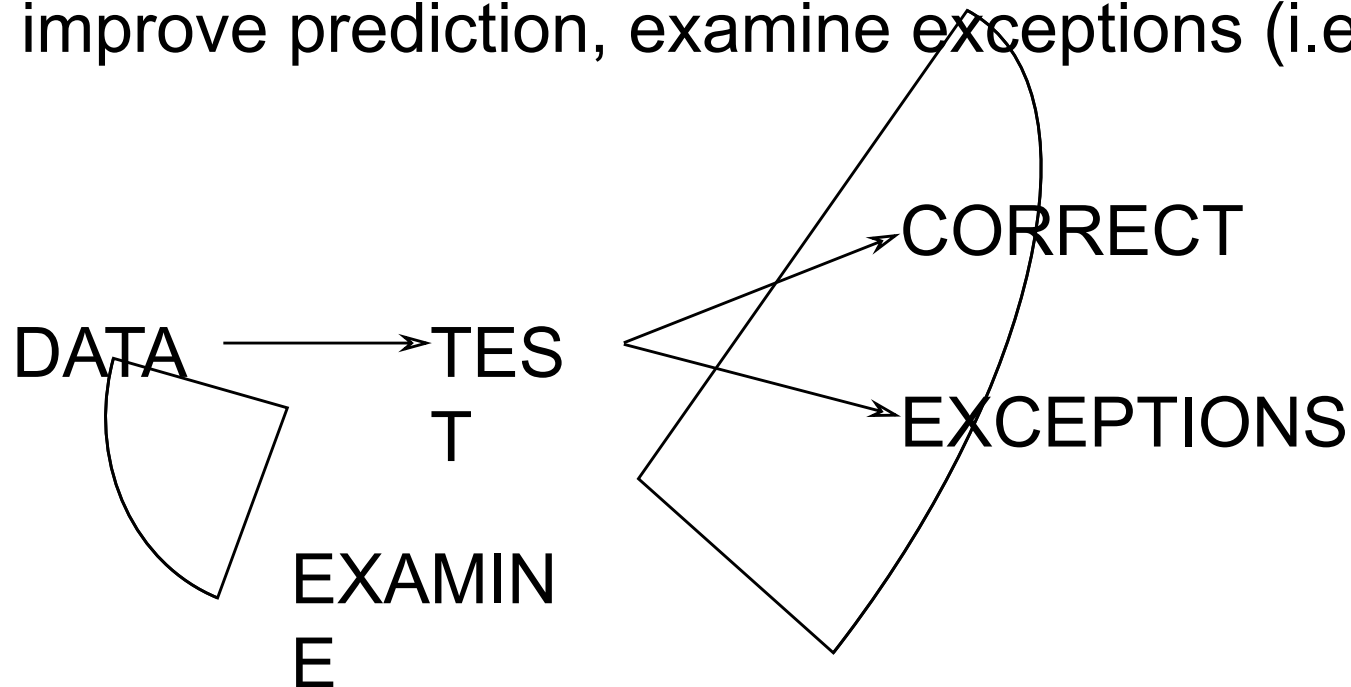


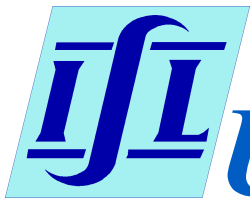
Why do you get complicated Rules?

- n Lack of truly discriminating attributes
revisit data
- n Simplistic approach to numeric attributes
add higher level derived values
- n "Special" cases
pruning (generality)
- n Poor coverage (overall or in subsets)
revisit data e.g. Balance
Try pruning the tree, group symbolic
- n A complex rule is often still *highly accurate*

Improving Performance

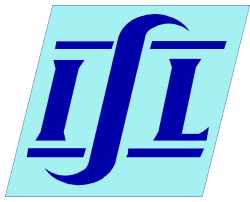
To improve prediction, examine exceptions (i.e errors)



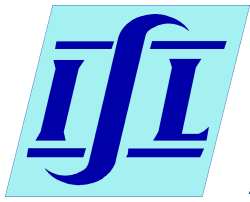


Using Induction to Investigate Performance

- n Induce rule using:
 - | test data
 - | rule's/net's decisions as additional inputs
- n If rule and net perform well, their decisions will be highly significant and hence near the top of the tree
- n Sub-trees describe error cases
 -
 - \$N-Risk Med Risk
 - \$R-Risk Med Risk
 - Age >= 28
 - Conviction F
 - Age < 30 -> **Low Risk**

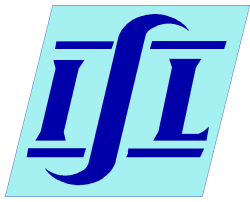


Advanced Features



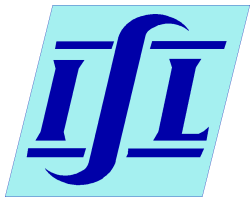
Manipulation nodes not covered

- n Aggregate
 - l summarises data
- n Sample
 - l extracts working sets
- n Balance
 - l compensates for skewed data
- n Sort
 - l sorts on one or more keys
- n Distinct
 - l handles duplicate records



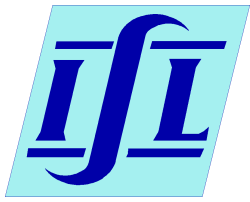
Handling Noisy Data

- n Data quality node
 - I interactive report on field completeness
- n “Blank” handling
 - I definition in type nodes
 - I repair by Filler node
- n Techniques for working with noisy data



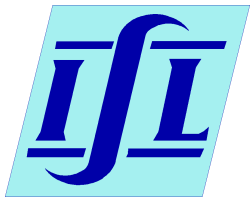
Advanced Modelling

- n Data preparation / transformations
- n Scoring models
- n Meta models
- n Confidence figures
- n “Expert” options on modelling nodes



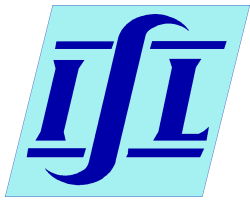
Time Series (1)

- n Special display types
 - l line plot, Multiplot node
- n History node to incorporate previous values
- n Transformation of “raw” point values to descriptions of change
- n Special Derive node types



Time Series (2)

- n CLEM sequence functions:
 - | @AVE, @SUM, @MAX, @MIN, @SDEV
 - | @DIFF1, @DIFF2
 - | @OFFSET, @SINCE, @SINCE0
- n Combination of dynamic/static data
 - | summarisation
- n Deployment issues



Other Topics

- n Caching
- n Reporting
- n Handling times and dates

